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Artificial Intelligence in Power Electronics

A perspective on opportunities, challenges, and future directions

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Authors and affiliations

The list is ordered alphabetically by family name.

Dr. Davide Barlini – Mitsubishi Electric R&D Centre Europe, FR
Vasilij Baumann – InstruNEXT, DE (SG 3 Chair)
Prof. Dr. Frede Blaabjerg – Aalborg University, DK
Jannes Briese – TU Braunschweig, DE
Dr. Raul Carreira Rufato – Safran Tech, FR
Dr. Ilknur Colak – Schneider Electric, FR
Johannes Drechsel – Fraunhofer IKTS, DE
Prof. Dr. Uwe W. Drofenik – Technical University of Vienna, AT (SG 1 Chair)
Dr. Maximilian Götzinger – Schneider Electric, AT
Prof. Dr. Norbert Grass – Ohm Univ. of Applied Sciences Nuremberg, DE
Thomas Harder – ECPE, DE (Task Force Coordination)
Prof. Dr. Rolf Hellinger – Siemens, DE
Kevin Hermanns – PE Systems, DE (SG 2 Chair)
Leonhard Hertenstein – Mercedes Benz, DE
Ingmar Kallfass – University of Stuttgart, DE
Prof. Dr. Nando Kaminski – Univ. of Bremen, DE
Robert Keilmann – TU Braunschweig, DE
Felix Keppner – Siemens, DE
Dr. Mohammed Ali Khan – SDU Sønderborg, DK (SG 4 Chair)
Prof. Dr. Johann W. Kolar – ETH Zurich, CH (SG 1 Chair)
Dr. Ivana Kovacevic-Badstuebner – ETH Zurich, CH
Dr. Varaha S. Bharath Kurukuru – SAL Silicon Austria Labs, AT (SG 4 Chair)
Prof. Dr. Leo Lorenz – ECPE, DE
Dr. Ingo Luedke – CSA Catapult, UK
Prof. Dr. Paolo Mattavelli – University of Padova, IT
Sven Meier – FAPS/Friedrich-Alexander Univ. Erlangen-Nuremberg, DE (SG 3 Chair)
Dr. Anton Miric – Heraeus Electronics, DE
Dr. Christoph Raab – Siemens, DE
Dr. Lars Rebenklau – Fraunhofer IKTS, DE
Dr. Claudio Rivetta – Inst. Tecnológico de Aragón, ES
Dr. Andreas Rosskopf – Fraunhofer IISB, DE
Prof. Dr. Subham Sahoo – Aalborg University, DK
Dr. Yanfeng Shen – Semikron Danfoss, DE
Dr. Nima Tashakor – Volkswagen, DE (SG 2 Chair)
Ali Tfaily – Schneider Electric, FR
Dr. Muhammad Usama – Mitsubishi Electric R&D Centre Europe, FR
Prof. Dr. Oliver Wallscheid – University of Siegen, DE (SG 4 Chair)
Prof. Dr. Huai Wang – Aalborg University, DK
Prof. Dr. Shuai Zhao – Aalborg University, DK

Corresponding organization: ECPE European Center for Power Electronics e.V., Nuremberg, Germany.

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Preface

Artificial intelligence (AI) is a hot topic these days, and there is certainly a lot of hype surrounding it, but it is bringing about lasting changes to the way we live and work. That is why we in the ECPE network have been asking ourselves what impact the use of AI is already having in power electronics (PE) today, and what further potential lies in future AI applications.

With this ECPE white paper, we aim to raise awareness of this topic, provide a platform for the discussion within the PE community, and strengthen collaboration between industry and academia.

What is PE? PE is an important subfield of electrical and electronics engineering dealing with the application of electronics to the control and conversion of electric power. PE is a cross-sectional and ubiquitous field as it covers many disciplines, including material science, semiconductor physics and power devices, assembly and interconnection technologies, electrical circuit topologies, and control, in all systems and applications dealing with electric energy from the low milliwatt range up to gigawatt power.

Why AI in PE? The key potential for the future in the use of AI in PE lies in speed – and consequently in cost. This includes shortening development and testing times, thereby reducing time-to-market; optimizing production by reducing failures and rejects; and intelligent condition monitoring and service life monitoring of products in the field.

The fundamental change associated with the use of AI in PE lies in the possibility that AI could fully automate the engineering process – in other words, the vision that AI will take full charge of the design, optimization, manufacture and testing of a power module or a converter. This will enable AI to enter the next level of cost reduction, but what will become of engineers' jobs? Our vision – or hope – is an AI-assisted engineer with outstanding work efficiency.

What does AI do? AI is a branch of computer science focused on building systems capable of performing tasks that typically require human intelligence. These tasks include learning, reasoning, problem-solving, understanding language, and recognizing patterns. AI operates by processing massive amounts of data through smart algorithms. Instead of just following a strict set of pre-written rules, the system identifies patterns in the data and "learns" from them over time to make decisions, predict outcomes, or generate new content.

This ECPE white paper focuses on the use of AI methods and tools in power electronics, rather than on the role of power electronics in AI data centres – which is also an interesting and highly relevant topic. Following an introduction setting out the fundamentals and terminology, prepared by sub-group (SG) 1, the structure of the white paper traces the application of AI methods and tools throughout the entire life cycle of a power electronics product – from the early development phase, through to manufacturing and finally the product's operational phase (SG2 to SG4).

The cross-cutting issue of the use of AI methods and tools in human learning and advanced training is covered in a separate document prepared by SG5.

We thank the authors for their support and for their valuable contributions to this ECPE white paper which highlights the role and importance of AI in PE throughout the entire product life cycle – from the early development phase right through to products in operation.

Prof. Dr. Leo Lorenz, ECPE President
Dipl.-Phys. Thomas Harder, ECPE General Manager

1 Introduction

AI is steadily gaining importance in power electronics [1, 2], opening new paths for converter optimization, accelerated design cycles, and intelligent condition monitoring. For electrical engineers, the topic can seem abstract, partly because AI originated in computer science rather than in classical engineering [3]. A practical, application-oriented perspective is, therefore, helpful when first entering this field.

In the following, we explore AI through three lenses that directly relate to the tasks and workflows power electronics engineers know well:

1. Pattern recognition using machine learning (ML) and deep neural networks (DNNs) [4, 5] to build efficient simulation models and perform failure analysis;

2. large language models (LLMs) that support design, analysis, and documentation;

1. reinforcement learning (RL) [6], which offers the possibility of discovering novel control strategies or converter concepts beyond conventional engineering knowledge and/or intuition.

First, we need to clarify the terminology. AI serves as the broad umbrella term for computational techniques that emulate aspects of human intelligence. ML is a subset of AI focused on systems that learn from data rather than being explicitly programmed. Inspired by the structure of the human brain, DNNs are machine learning models composed of many layers of artificial neurons – an input layer, one or multiple hidden layers, and an output layer, where the neurons in the hidden layers compute a weighted sum of the neurons from the previous layers and apply a nonlinear activation function to that sum (this activation function allows the DNN to learn nonlinear relationships in the data) – that enable them to extract complex features from high-dimensional data. LLMs represent a newer category of AI trained on huge amounts of text to assist with reasoning, explanation, and design-related communication. RL describes systems that learn by interacting with an environment and receiving feedback in the form of rewards (learning via "trial and error"). RL does not need labeled data but accumulates information during operation by interacting with an environment. The relationship between the methods discussed above is shown in Figure 1.1. Although these definitions may appear formal, they will naturally guide us toward practical engineering roles.

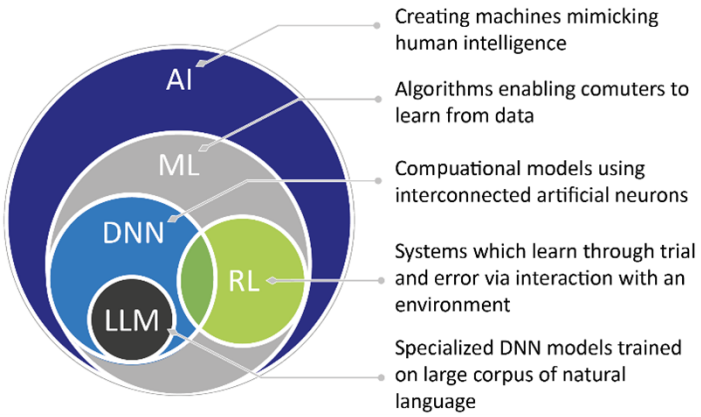


Figure 1.1: The field of AI attempts to create systems that mimic human intelligence. The diagram shows how the terminologies AI, ML, DNN, LLM, and RL relate to each other.

1.1 Pattern Recognition with Machine Learning and Deep Neural Networks

For engineers familiar with classical modeling, it is helpful to view machine learning as an extension of curve-fitting. Linear regression, a tool every engineer has already used, fits a line to data to approximate behavior, e.g., estimating switching losses from switched current (at given voltage) or determining conduction losses as a function of temperature. ML builds on this idea but scales it massively. Instead of fitting a simple curve, it finds relationships in datasets with thousands or even millions of input variables. In this context, linear regression, which is simple linear curve fitting, is an ML algorithm, and is, therefore, already by general definition a form of AI.

A simple illustration of very-high-order curve fitting is an image of 100 by 100 pixels, which results in 10'000 input parameters (dimension is 10'000). Predicting a physical quantity - for instance, an inductance value derived from a magnetic field simulation image [7] – becomes a task with enormous dimensionality. No engineer could write explicit rules for such mapping, yet an ML algorithm like a DNN can learn it autonomously. By stacking many (hidden) layers or artificial neurons, each extracting increasingly meaningful features, a DNN learns to generalize images rather than memorize them. This is fundamentally different from approaches such as lookup tables, which require storing every possible case and cannot extrapolate beyond the known data.

In the example from above, the mathematical process of "DNN training is learning from data" is employing a DNN algorithm from an ML library and feeding several thousand images containing different geometry drawings (e.g., core, air gap, windings of an inductor or transformer) plus the finite element method (FEM)-calculated inductance associated with the geometry shown in each image [7]. This requires substantial computational resources, but can use existing open-source software. After this training, the DNN can predict the inductance of new geometry drawings if it is not too different from the geometry drawings employed during training.

In power electronics, this ability is extremely valuable. The field of power electronics generates complex and diverse data like electromagnetic field distributions from FEM tools, temperature- and flow-data from computational fluid dynamics (CFD) simulations, measured switching waveforms, electromagnetic interference (EMI) spectra, acoustic noise, visual inspection images of PCBs and packaging structures, extensive semiconductor characterization, and other device data. DNNs trained on such information can serve as *fast surrogate models* [7] *that approximate physics-based simulations* with remarkable speed and accuracy. They enable engineers to build models that allow them to explore component choices, layouts, cooling strategies, and switching conditions orders of magnitude faster than conventional simulations would permit.

Machine learning is also used in advancing *health monitoring* [8]. By evaluating images of semiconductor modules, acoustic patterns from mechanical stress, or electrical signals captured during switching transients, ML models can detect early signs of degradation such as bond-wire lift-off, solder voiding, or thermal fatigue. This capability leads to predictive maintenance and increased system reliability.

Another emerging area is *device modeling* [9, 10]. Manufacturers provide extensive datasheets and measurement curves, yet engineers still spend significant time extracting parameters and constructing simulation models. DNNs trained on empirical semiconductor data can capture nonlinearities and temperature effects more accurately than analytical equations, while remaining computationally efficient.

In summary, ML enhances the engineer's toolbox [11]. It learns from simulation and measurement data, supports faster, more accurate modeling, and provides new diagnostic capabilities, ideally complementing rather than replacing established physical understanding.

1.2 Large Language Models as Design Assistants

DNNs help recognize patterns, and LLMs assist with knowledge processing and engineering workflows [12, 13, 14]. Trained on vast amounts of technical text, LLMs support documentation, reasoning, coding, and literature interpretation. They are, in essence, extremely large DNNs - often with billions of parameters - designed to learn the statistical structure of language. During training, the DNN repeatedly predicts, based on statistics, the next word in a sentence and adjusts its internal parameters depending on whether it guessed correctly. Through this process, it captures grammar, technical terminology, logical relationships, and factual associations. At a certain size of the DNN, its answers seem intelligent and make sense in context, although they are mainly based on statistics ("which word follows two previous words with what probability?") and some randomization. The enormous size of LLMs enables them to store subtle patterns and connections across massive domains of knowledge, allowing them to generate coherent text, explain engineering concepts, and provide informed design assistance based on learned context rather than fixed programmed rules.

While DNNs identify patterns in numerical or image data, LLMs support a different category of engineering work, especially reasoning, documentation, and knowledge access. LLMs are trained on vast amounts of technical literature, code, and reports, enabling them to assist with tasks involving text, explanations, and structured problem-solving. Training LLMs, especially on power electronics data, e.g., related publications in the IEEE Xplore database, would create a very powerful virtual assistant for power electronics engineers.

Engineers can interact with such models conversationally, asking for help in understanding application notes, drafting design documentation, or generating simulation scripts in MATLAB, Python, or SPICE. They can also request guidance on highly specific topics such as gate-drive tuning for SiC devices, trade-offs between switching speed and electromagnetic interference, or cooling design considerations. One very powerful way to use LLMs in power electronics is to select components for a circuit based on current and voltage stress, which is usually a very time-consuming process (e.g., there are currently more than 7,000 different types of insulated-gate bipolar transistors (IGBTs) and IGBT modules in DigiKey's online database). The user provides a prompt, which is simply the instruction or question that guides the model's response. Well-written prompts will provide better guidance and result in significantly higher quality answers from the LLM.

Rather than replacing engineering judgement, LLMs act as quasi-intelligent assistants that provide quick access to knowledge and support exploration during the conceptual and early design phases [15, 16]. They are particularly useful when reviewing large volumes of information or when drafting structured analyses.

1.3 Reinforcement Learning for Exploratory Converter Development

LLMs help engineers use what is already known. RL, by contrast, has the potential to explore what is not yet known [17, 18]. Instead of learning from a fixed dataset, an RL agent interacts with a simulated environment, experimenting with control actions or structural parameters and receiving rewards for desirable behavior – such as high efficiency, low component stress, sinusoidal current control / low harmonic distortion (see Figure 1.2), or fast transient response. There, sinusoidal input current control of a single-phase full-bridge alternating current (AC) / direct current (DC) converter (environment) using a DNN parametrized by a deep reinforcement learning (DRL)

algorithm based on a reward function, i.e., the weighted sum of the deviation from the sinusoidal current reference and the switching losses, is shown.

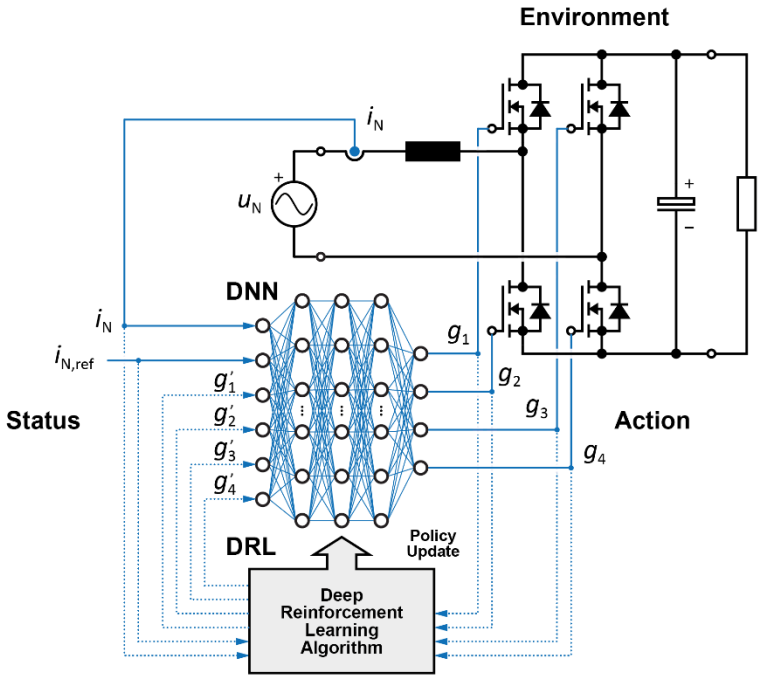


Figure 1.2: Sinusoidal input current control of a single-phase full-bridge AC/DC converter.

The DRL algorithm does not require user-provided training data; instead, it explores the topology, evaluates the success of control actions based on the reward, and finally defines the weights and activation functions of the DNN to maximize the reward ("learning by doing"). The DNN-based controller can even handle situations, such as load steps, that were never encountered during training [18]. With reference to the designation commonly used in the field of RL, the action (gate signals $g_i(t)$, $i = 1, 2, 3, \ldots$) on the environment (converter-circuit) is defined by the agent (DRL algorithm and DNN with three hidden layers and 64 neurons per layer) based on the state (current reference value i_N^*, input current i_N and gate signals of the last time step $g_i' = g_i(t-1)$).

This paradigm closely resembles how human engineers gain intuition, e.g., by trying arrangements of circuit elements, observing behavior, and iteratively refining decisions. However, unlike a human, an RL system can systematically explore a vast design space without being constrained by established topologies or control heuristics. As a result, RL may uncover unconventional modulation techniques, soft-switching strategies, or hybrid converter architectures that would not easily arise from traditional design thinking.

Although RL requires computational training environments and careful reward definition, its conceptual appeal is clear. It represents a path toward automated discovery — potentially leading to novel power-electronic design approaches.

1.4 Summary

AI is not an alternative to classical power electronics engineering, but it is a natural extension of it. When viewed through the practical lenses of pattern recognition, language-assisted design,

and RL-based exploration, AI becomes far less abstract and much more aligned with everyday engineering practices.

DNNs facilitate rapid modeling and health monitoring, while LLMs enhance engineering productivity and knowledge access. RL potentially unlocks new converter concepts and control strategies. Together, they form a set of tools that complement physics-based understanding and accelerate innovation in power electronics.

All in all, these AI technologies extend traditional power-electronics tools, supporting faster design cycles, improved reliability, and potentially creating new converter and control concepts. Up to now, the foundation remains solid electrical engineering, and AI adds a powerful new layer of capability.

2 Scope and Role of AI in Power-Electronics Designs

The increasing complexity of modern power-electronics systems – driven by wide bandgap (WBG) devices, stringent electromagnetic compatibility (EMC) requirements, strong multi-physics coupling, and accelerated development cycles – has pushed conventional design workflows to their limits. AI methods offer new means to augment and partially automate design, optimization, qualification, and validation across the power-electronics lifecycle. This chapter provides a structured and critical overview of AI methods and tools in power-electronics design and engineering, spanning concept exploration, component-level co-design, simulation acceleration, reliability and lifetime prediction, and design verification.

The chapter examines how AI can address key design trade-offs between cost, performance, and robustness, while highlighting challenges related to data availability, generalization, bias, and trustworthiness. AI-supported topology exploration is then addressed, including automated discovery as well as reinforcement learning as a potential enabler for topology evaluation, and the limitations imposed by high-dimensional design spaces. Furthermore, the chapter addresses multi-physics, AI-enhanced component-level co-design, including surrogate models for semiconductors, magnetic components, and passives; multi-objective optimization (MOO) of component parameters and layouts; and AI-accelerated simulation as an enabler for fast performance evaluation, reliability assessment, and lifetime prediction.

Finally, the outlook highlights the growing impact of LLMs and agentic workflows on future design processes, enabling automated requirement management, human-in-the-loop engineering, and certifiable AI toolchains. The chapter concludes with a call for coordinated action across industry, academia, and funding bodies to support the trustworthy and sustainable adoption of AI in power-electronics design and engineering.

The design of modern power-electronics systems is increasingly challenged by the simultaneous escalation of system complexity and development pressure. The widespread adoption of WBG semiconductor devices has significantly expanded the available design space, enabling higher switching frequencies, increased power density, and improved efficiency. At the same time, these advances intensify multi-physics coupling effects between electrical, thermal, and mechanical domains and exacerbate EMC constraints. Conventional design methodologies – largely based on sequential domain-specific optimization and manual iteration – are increasingly unable to cope with the resulting dimensionality and interdependencies.

In parallel, market and regulatory pressures demand shorter development cycles, higher product reliability, and improved cost–performance trade-offs. Design teams are therefore required to evaluate a growing number of hardware and control strategies under tighter time-to-market constraints, often with incomplete knowledge early in the design phase. This situation exposes fundamental limitations of traditional design workflows, particularly in the context of early-stage exploration and system-level optimization.

AI methods offer a complementary paradigm by enabling data-driven abstraction, fast surrogate modeling, and automated exploration of large and highly coupled design spaces. The objective is not to replace physical modeling or engineering expertise, but to augment existing workflows with tools capable of identifying non-intuitive solutions, accelerating design iterations, and systematically balancing conflicting objectives. In this context, the role of AI in power-electronics

design extends beyond pure performance optimization and also includes embedding reliability and lifetime considerations early in the design process, and enabling automated documentation and traceability across the design lifecycle.

2.1 Trade-Offs and Challenges

The adoption of AI in power-electronics design must be judged against the classical trade-off between cost, development time, and performance. AI can accelerate parameter sweeps, reduce manual iterations, and support exploration of non-intuitive solutions, but it also introduces effort for data generation, model training, tool integration, validation, and maintenance [19]. Therefore, AI is most useful where the design space is complex, conventional simulation or testing is expensive, and the expected performance gain justifies the additional overhead. For simple and well-understood design tasks, analytical models or conventional optimization may remain more efficient and transparent.

A major limitation is the availability and management of high-quality data. Power-electronics data are often fragmented across simulation tools, measurements, supplier information, and project-specific design files. Differences in operating conditions, device generations, thermal boundaries, modulation strategies, and layout parasitics reduce reusability and comparability. Without structured metadata, version control, and traceable data pipelines, AI models may learn test-setup artifacts rather than physically meaningful design relations [20].

Generality and trustworthiness remain critical challenges. A model trained for one topology, voltage class, semiconductor technology, cooling concept, or modulation strategy may not extrapolate reliably to another design family. Training data may also overrepresent nominal or successful designs while underrepresenting rare faults, tolerances, aging effects, and worst-case conditions. Trustworthy AI therefore requires uncertainty quantification, explainability, robustness checks, and validation against independent simulations or measurements, particularly for safety-relevant applications.

Finally, power electronics lacks widely accepted public benchmark datasets for AI-assisted design tasks such as topology selection, loss prediction, EMI-aware layout optimization, thermal co-design, and lifetime estimation. This makes fair comparison and reproducibility difficult. Standardized reference converters, operating profiles, datasets, and evaluation metrics are needed to assess not only prediction accuracy, but also extrapolation capability, computational cost, physical consistency, and integration effort in realistic engineering workflows. This becomes even more important as LLM-based and agentic design tools enter power-electronics workflows [21].

2.1.1 Decision Matrix for AI Potential in a Design Application

Incorporating AI into power-electronics design requires a structured evaluation of where AI can provide measurable engineering value. AI should not be introduced only because a task is technically possible to automate. If the design problem is simple, analytically well understood, or inexpensive to evaluate, conventional models and optimization methods may remain more transparent and efficient. Conversely, AI becomes attractive when the design space is high-dimensional, nonlinear, multi-objective, costly to evaluate, or only partially captured by analytical models. This is consistent with multi-criteria decision analysis, where alternatives should be evaluated through explicit criteria, weights, and trade-offs rather than by a single qualitative judgment [19, 22].

For power electronics, the decision should distinguish three aspects: *AI benefit potential*, *AI feasibility*, and *deployment risk*. Benefit potential describes whether AI is likely to improve the engineering workflow; feasibility describes whether a reliable model can realistically be developed; and deployment risk describes whether the model can be used safe and traceable in the intended application. Table 2.1 summarizes the proposed screening matrix. Each criterion can be scored as low, medium, and high, with a certain level of qualitative flexibility.

Table 2.1: Screening Matrix for AI Potential in Power-Electronics Design (Each Criterion can be Scored as Low, Medium, and High, with a Certain Level of Qualitative Flexibility Using an Integer Number Between 0 and 10)

	Criterion	Low (0-3)	Medium (4-7)	High (8-10)	Meaning of a high score
Benefit	Search-space complexity	Few weakly coupled variables	Several coupled parameters	High-dimensional, nonlinear, mixed discrete-continuous space	AI may improve exploration
	Model uncertainty	Reliable analytical model exists	Approximate model with known limits	Strong empirical dependence or no accurate analytical model	AI may capture missing relations
	Multi-objective coupling	Objectives mostly independent	Moderate trade-offs	Strong Pareto conflict between efficiency, EMI, cost, thermal stress, volume, or lifetime	AI/MOO may improve trade-off search
	Iteration cost	Fast analytical calculation	Moderate simulation effort	Expensive finite element analysis (FEA), CFD, EMC simulation, mission-profile run, or lab test	Surrogates can reduce evaluation cost
	Performance sensitivity	Minor local impact	Relevant component-level impact	Strong effect on system KPIs such as efficiency, EMI, power density, cost, or lifetime	Optimization has high value
Feasibility	Data availability and quality	Sparse, inconsistent, poorly labelled	Partial operating-space coverage or incomplete validation (e.g., simulation-only)	Structured, representative, traceable simulation/measurement data	Model training and validation are realistic
	Domain complexity	Single physical domain	Two coupled domains	Electrical, thermal, magnetic, EMI, mechanical, and aging coupling	AI can support coupled abstraction
	Model dimensionality	Scalar or 1D features	Waveforms, maps, or 2D fields	3D fields, graphs, layouts, or time-dependent multi-domain data	Advanced AI architectures may be justified

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	Criterion	Low (0-3)	Medium (4-7)	High (8-10)	Meaning of a high score
	Computational resources	Severe compute/memory limits	Workstation or edge constraints	Offline CPU/GPU/high-performance computing (HPC) resources available	Larger models or training campaigns are feasible
Risk	Criticality level	Non-critical design support	Engineering decision support	Safety-critical or qualification-relevant decision	Black-box autonomy is restricted
	Regulatory burden	No formal compliance relevance	Internal process relevance	International Electrotechnical Commission (IEC), International Organization for Standardization (ISO), UL, or automotive qualification relevance	Traceability and V&V dominate deployment
	Stringent latency requirement	Offline use	Near-real-time engineering support	Embedded or real-time control/protection	Model size and determinism are constrained
	Interpretability requirement	Low	Medium	High	Physics-informed or transparent models are preferred
	Cost of wrong prediction	Low consequence	Correctable engineering error	Hardware damage, safety risk, or invalid qualification result	Independent validation is mandatory

Instead of collapsing all criteria into one score, three normalized scores are calculated:

$$B = \frac{\sum_i w_i b_i}{10 \sum_i w_i}, \qquad F = \frac{\sum_k u_k f_k}{10 \sum_k u_k}, \qquad R = \frac{\sum_j v_j r_j}{10 \sum_j v_j} \tag{2.1}$$

where B is the AI benefit potential, F is the AI feasibility, and R is the deployment risk. The terms b_i, f_k, and r_j are the criterion scores, while w_i, u_k, and v_j are application-dependent weights. The weights should be selected by domain experts; if no priority is known, equal weights can be used as a first screening assumption. The purpose of the score is not to provide a universal truth, but to make the decision transparent and reproducible. The resulting classification can be interpreted as given in Table 2.2.

This separation is important because a high-risk application is not automatically a poor AI use case. For example, a safety-relevant protection function may have high benefit potential, but the acceptable AI role may be limited to offline design exploration, anomaly screening, test-case generation, or decision support. It should not be deployed as an unconstrained black-box real-time controller without deterministic fallback and formal validation.

As sanity checks, EMI-aware layout optimization and magnetic-loss surrogate modeling under arbitrary waveforms would typically score high in benefit because they involve nonlinear, multi-domain, and expensive-to-evaluate design spaces. They may also be feasible if simulation or

Table 2.2: Interpretation of Normalized Screening Scores

Condition	Interpretation
$B \geq 0.6$, $F \geq 0.6$, and $R \leq 0.4$	Strong AI candidate; AI is likely justified for design automation, surrogate modeling, or optimization
$B \geq 0.6$, and neither the strong nor research condition applies	Conditional AI candidate; use only with physics constraints, uncertainty quantification, traceability, and human-in-the-loop validation
$B \geq 0.6$ and $\big((F < 0.4$ and $R \leq 0.4)$ or $(F \geq 0.6$ and $R > 0.6)\big)$	Research candidate; the use case is promising, but data, validation, or tooling are insufficient for deployment, hence industry adaptation may therefore be limited
$B < 0.6$	Weak AI candidate; analytical models, rule-based methods, or conventional optimization are likely preferable
High criticality + high latency requirement + low interpretability	Not suitable as a standalone black-box AI function, independent of the benefit score

measurement data are structured and independently validated. In contrast, simple capacitor sizing from a known ripple-current equation would score low in benefit because the analytical model is transparent, inexpensive, and already adequate. A safety-critical real-time protection task may score high in benefit but also high in deployment risk; therefore, the matrix would classify it as a conditional or non-standalone AI application rather than a direct black-box deployment.

This decision matrix should therefore be understood as a practical screening and prioritization tool. It supports value-driven AI adoption by identifying cases where high design complexity, incomplete models, costly iterations, and strong multi-physics coupling coincide with sufficient data quality and a realistic validation path. Such a structured approach is consistent with the broader view that AI in power electronics should support design, control, maintenance, optimization, regression, classification, and data-structure exploration, while remaining grounded in physical constraints and engineering verification [23, 24].

2.1.2 Prevalent AI Architectures and Models in Design Applications

Given the strong nonlinearities, multi-physics coupling, limited labelled data, and strict reliability requirements in power-electronics design, model selection should be driven by the task and deployment context rather than by the popularity of a specific architecture. In practice, AI models are rarely used as standalone black boxes. They are embedded into physics-based simulation, optimization, verification, or human-in-the-loop design workflows. Table 2.3 separates learning paradigms, model families, and design roles, while Table 2.4 gives indicative data and compute requirements.

Table 2.3: AI Model Families Relevant for Power-Electronics Design

Model family / paradigm	Representative methods	Typical use in power electronics	Strengths	Main limitations	Refs.
Classical supervised learning	Linear/kernel regression, support vector machines, random forests, gradient boosting	Fast loss, efficiency, temperature, feasibility, and fault classifiers/estimators; early-stage surrogate models	Works with moderate datasets; CPU-friendly; relatively interpretable	Limited extrapolation; feature engineering often required; weak for high-dimensional fields or layouts	[19], [22]
Deep supervised surrogates	Feedforward neural networks, convolutional neural networks (CNNs), multi-output neural networks, residual networks	Converter and component surrogate models; fast prediction of losses, thermal stress, EMC indicators, lifetime metrics	Fast inference; strong nonlinear approximation; suitable for optimization loops	Needs labelled simulation/measurement data; can overfit; uncertainty is usually not native	[19, 22, 25]
Temporal and sequence models	recurrent neural networks (RNNs), long short-term memorys (LSTMs), gated recurrent units (GRUs), temporal convolutional networks, transformer encoders	Mission-profile modeling, degradation and aging prediction, waveform-based diagnostics, dynamic converter modeling	Captures time dependence and waveform history	Training can be data-intensive; long-horizon stability and extrapolation remain difficult	[19, 20]
Unsupervised and representation learning	principal component analysis (PCA), uniform manifold approximation and projection (UMAP), clustering, autoencoders, self-supervised encoders	Design-space visualization, operating-point clustering, anomaly discovery, feature compression before optimization	Useful when labels are scarce; improves data understanding and preprocessing	Does not directly provide performance prediction; latent variables may lack physical meaning	[19], [22]
Probabilistic and active-learning surrogates	Gaussian processes, Bayesian neural networks, polynomial chaos expansion, Bayesian optimization, active learning	Uncertainty-aware design, tolerance studies, expensive FEA/CFD/EMC optimization, adaptive sampling	Sample-efficient; uncertainty-aware; suitable when each simulation or test is expensive	Gaussian processes scale poorly with dataset size; uncertainty quality depends on assumptions	[22], [25]
Reinforcement learning	DQN, proximal policy optimization (PPO), soft actor-critic (SAC), model-based RL, constrained RL	Modulation/control tuning, topology-control co-design, sequential parameter decisions	Learns sequential policies; can discover non-intuitive control strategies; relevant for topology feasibility	High sample cost; reward design is difficult; safety and stability are not guaranteed	[23]

Continued on next page

Table 2.3 continued from previous page

Model family / paradigm	Representative methods	Typical use in power electronics	Strengths	Main limitations	Refs.
Physics-informed and hybrid models	physics-informed neural networks (PINNs), physics-informed DeepONets, Fourier neural operators, neural operators, ML-corrected reduced-order models	Fast field prediction, inverse design, coupled electrical–thermal–magnetic simulation, mission-profile studies	Embeds physical constraints; can reduce dependence on labelled measurements; suitable for parametric partial differential equation (PDE)-type problems	Training is difficult; sensitive to scaling and loss weighting; industrial robustness still limited	[26, 27]
Graph-based models	Graph neural networks, graph transformers, circuit graph encoders	Circuit topology representation, layout analysis, parasitic modeling, topology discovery	Natural representation of circuits and layouts; handles variable-size structures	Requires graph-labelled or simulation-generated circuit/layout datasets; explainability and validation remain challenging	[24]
Generative and foundation models	Transformers, diffusion models, multimodal foundation models, large language models	Requirement analysis, script/code generation, simulation orchestration, design assistance, candidate topology/layout generation	Strong representation learning; useful for automation, documentation, and tool orchestration	Hallucination risk; expensive to train or fine-tune; immature for safety-critical hardware decisions	[28, 29]
Interpretable emerging models	Kolmogorov–Arnold networks, symbolic regression, sparse identification	Adaptive control, compact surrogate models, interpretable function approximation	Potentially more interpretable than dense neural networks; compact models possible	Early-stage in power electronics; robustness and benchmarking still limited	[30]

The values in Table 2.4 are order-of-magnitude screening values, not universal requirements. The required dataset size depends strongly on dimensionality, noise level, operating-space coverage, and whether data come from measurement, high-fidelity simulation, or synthetic design-of-experiments campaigns.

Table 2.4: Indicative Data and Compute Envelope for Model Selection

Model family / paradigm	Training data or sampling effort	Training compute	Inference footprint	Practical maturity in PE design	Suitable deployment role
Classical supervised learning	10^2–10^4 labelled samples	CPU; minutes to hours	kB–MB; μs–ms	High for diagnostics and low-dimensional surrogates	Offline and embedded support
Deep supervised surrogates	10^3–10^6 labelled simulation/ measurement samples	GPU useful; hours to days	MB–100 MB; μs–ms	Medium to high for simulation-trained surrogates	Offline optimization, fast design screening

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Table 2.4 continued from previous page

Model family / paradigm	Training data or sampling effort	Training compute	Inference footprint	Practical maturity in PE design	Suitable deployment role
Temporal and sequence models	10^3–10^6 labelled or self-supervised sequences/ mission profiles	GPU usually required; hours to days	MB–100 MB; ms	Medium for monitoring/ control-oriented tasks	Monitoring, dynamic modeling, offline design
Unsupervised/ representation learning	10^3–10^6 unlabelled samples	CPU/GPU depending on model	MB–100 MB; ms	Medium for data analysis and preprocessing	Data exploration, feature extraction
Probabilistic and active-learning surrogates	10^2–10^4 expensive evaluations initially	CPU/GPU; cost grows strongly with sample count	MB-scale; ms–s	Medium for expensive simulations	Adaptive sampling, Bayesian optimization
Reinforcement learning	10^5–10^8 environment steps in many deep-RL cases; lower for simple/ model-based cases	GPU/HPC often required; days possible	MB–100 MB; μs–ms after training	Medium for control; early for topology design	Offline training, constrained control studies
Physics-informed/ neural-operator models	10^4–10^7 collocation points or simulation snapshots; not necessarily labelled measurements	GPU/HPC often required; hours to days/weeks	MB–GB; ms–s	Early to medium for PE-specific multi-physics	Fast field solvers, inverse design, surrogate PDE models
Graph-based models	10^3–10^6 simulated or measured circuits/layouts/ graphs	GPU usually required	MB–100 MB; ms	Early but promising for topology/layout	Topology exploration, layout/parasitic screening
Generative/ foundation/LLM agents	Fine-tuning: 10^3–10^5 domain examples; pretraining: much larger	Large GPU clusters for training; API/ local GPU for use	GB-scale for local models; external if API-based	Early for hardware design, higher for code/ documentation	Human-in-the-loop design assistance
Interpretable compact models	10^2–10^4 samples or sparse experimental data	CPU/GPU; minutes to hours	kB–MB; μs–ms	Early stage in power electronics	Adaptive control, transparent surrogates

The table highlights that no single model class dominates all power-electronics design tasks. Classical machine learning and gradient boosting remain attractive for low-dimensional and tabular problems. Deep neural surrogates become useful when nonlinear mappings must be evaluated repeatedly. Probabilistic models and active learning are valuable when each simulation or experiment is expensive. Reinforcement learning is most relevant for sequential design and control decisions, particularly when topology evaluation requires a viable control strategy. Physics-informed and graph-based models are promising for multi-physics fields, layouts, and topology discovery, but require careful validation. LLMs and agentic workflows are currently strongest for requirement analysis, documentation, code/script generation, and simulation orchestration; they should not be treated as autonomous authorities for safety-critical hardware decisions.

2.2 AI-Supported Topology and Architecture Exploration

2.2.1 Automated Topology Discovery

Automated topology discovery aims to generate and evaluate novel converter structures beyond established design templates. Recent approaches leverage graph-based representations (e.g., circuit graphs) combined with search strategies such as evolutionary algorithms, graph neural networks (GNNs), and generative models to propose candidate topologies. In practice, the exploration space is vast and highly combinatorial, requiring *constraint-aware pruning* to remain tractable. Hard constraints (e.g., voltage/current limits, device ratings) and soft constraints (e.g., efficiency targets, EMC margins, thermal limits, manufacturability) are typically embedded as feasibility filters or incorporated into objective functions to discard infeasible designs early and focus the search on physically meaningful regions.

A critical component of automated discovery is *benchmarking and ranking* of candidate topologies. This involves consistent evaluation across standardized operating points and metrics (e.g., efficiency maps, loss breakdown, thermal headroom, component count, cost proxies, EMC indicators). Surrogate models and reduced-order simulations are often used to enable high-throughput screening, followed by selective high-fidelity validation. However, benchmarking remains sensitive to modeling assumptions and may bias results toward known architectures if constraints or priors are not carefully defined.

The primary challenge is *verification of newly generated topologies*. Unlike parameter optimization of known designs, novel structures lack established control strategies and validated operating modes. Without a suitable control policy, performance metrics cannot be reliably assessed, leading to incomplete or misleading evaluations. This limitation motivates the integration of control synthesis into the discovery loop – most notably via reinforcement learning or generative AI – which can infer viable control strategies for candidate topologies and thus enable meaningful, closed-loop evaluation.

2.2.2 Reinforcement Learning as a Key Enabler for Automated Topology Exploration

RL is relevant for power-electronics design because many design problems are sequential, nonlinear, constrained, and partly discrete. Unlike supervised learning, RL does not require labelled input–output pairs for every operating point. Instead, an agent interacts with a simulation model, digital twin, or experimental environment, observes a state s_t, selects an action a_t, and updates its policy $\pi\left(a_t|s_t\right)$ to maximize the expected cumulative reward:

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T} \gamma^t r_t \right]. \tag{2.2}$$

Here, r_t denotes the scalar reward at time step t, T is the finite episode horizon, $\gamma \in [0, 1]$ is the discount factor, and $\mathbb{E}_\pi[\cdot]$ denotes the expectation over trajectories generated by policy π and the environment dynamics. The policy is updated during training to maximize $J(\pi)$; in offline-trained engineering applications, the validated policy can subsequently remain fixed during deployment. The relevance of RL for design extends beyond conventional controller tuning. It can support design automation at four levels. First, pre-trained policies can provide fast tuning of controller parameters, modulation variables, or operating points. Second, RL can replace or augment heuristic optimizers, such as genetic algorithms, in high-dimensional searches where gradients are unavailable or unreliable. Third, RL can support topology exploration by treating switching states, connection choices,

component activation, or symbolic graph transformations as actions. Fourth, and most importantly for automated topology discovery, RL can evaluate whether a newly generated converter structure can actually be operated by identifying a feasible modulation or control policy, as illustrated in Figure 2.1.

This last role is the key distinction. A topology generator may produce many circuits that are structurally valid, but structural validity alone does not prove engineering usefulness. A candidate converter must regulate the required variables, respect device and thermal limits, avoid unsafe switching states, and operate across the intended source and load range. RL links structural exploration with closed-loop behavior: for each candidate topology, the agent searches for an operating policy and receives rewards or penalties based on the resulting dynamic performance. In this sense, RL acts as a feasibility filter before the topology is promoted to high-fidelity simulation, detailed optimization, or experimental validation.

Table 2.5: RL Roles, Action Spaces, Values, and Limitations in Converter Design

RL role in design	Design variable / action space	Main value	Main limitation
Performance tuning	Controller gains, modulation variables, operating setpoints	Fast adaptation of known systems	Stability and robustness must be verified
Optimization support	Initial candidates, search direction, exploration strategy	Can accelerate heuristic multi-objective searches	No guarantee of global optimality
Control synthesis	Switching states, duty cycles, modulation patterns	Determines whether a candidate topology can be operated	High simulation and sample cost
Topology generation	Graph actions, symbolic transformations, component insertion/removal	Enables exploration beyond predefined templates	Requires strict physical and graph-validity constraints

Recent work supports this interpretation. AutoTG formulates converter design as a symbolic optimization problem and uses RL to generate candidate power-converter topologies according to user specifications and preferences [31]. Earlier work on automatic converter design from specifications demonstrated RL-based topology-space exploration [32]. Recent reviews further classify RL applications in power converters into topology derivation, parameter design, and control implementation [23], while converter-control-focused reviews summarize the corresponding control applications and implementation challenges [33].

Despite this potential, RL-based topology exploration remains constrained by reward definition, sample complexity, and validation effort. The reward must combine conflicting objectives such as efficiency, tracking performance, thermal stress, EMC behavior, device utilization, cost, and robustness without encouraging physically invalid or simulation-specific solutions. Consequently, RL should be used as part of a hybrid design workflow with graph-validity checks, physics-based constraints, surrogate models, uncertainty estimation, and human-in-the-loop validation.

2.2.3 Dimensionality Problem

A fundamental limitation of AI-based design methods in power electronics arises from the extreme dimensionality of realistic design spaces. Converter architectures, component parameters, control variables, and operating conditions collectively result in a high-dimensional, mixed discrete–continuous optimization problem. While AI methods are often effective at identifying local optima, guaranteeing proximity to global optima remains challenging, particularly when data is sparse

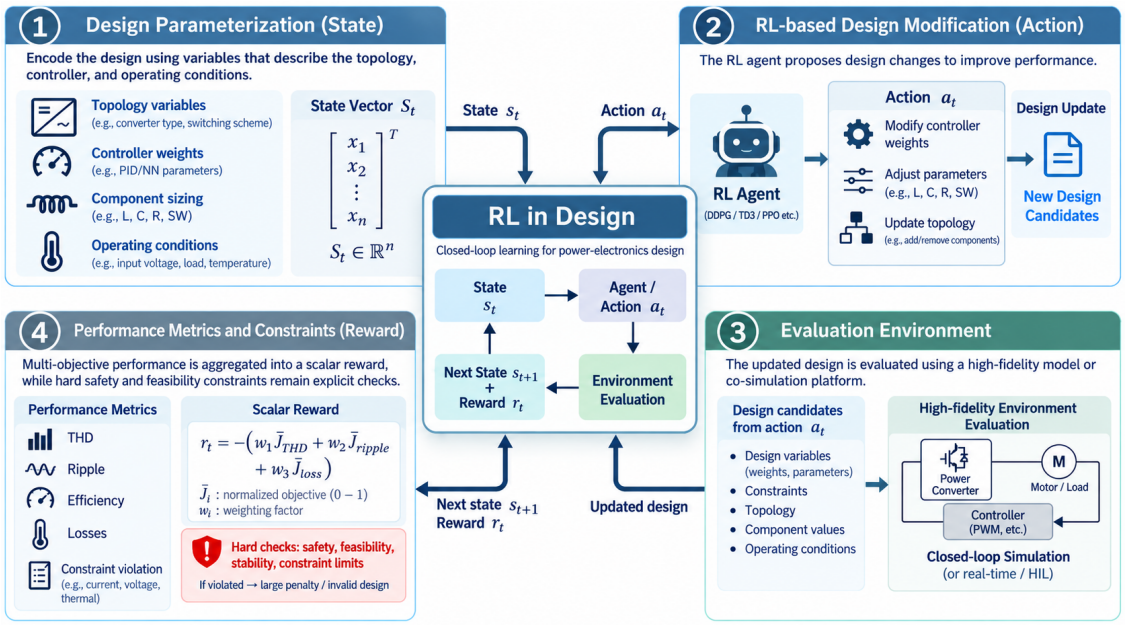


Figure 2.1: RL-based design optimization, where parameterized system states are updated through learned actions, evaluated in a simulation environment, and optimized via reward signals reflecting system performance.

or computational budgets are limited. Brute-force exploration is infeasible, and unconstrained learning risks converging to solutions that are optimal only within narrow regions of the design space. This dimensionality issue highlights the necessity of incorporating physics-based constraints, hierarchical design decomposition, and informed reduction of degrees of freedom to make AI-driven exploration tractable and engineering-relevant. Sensitivity analysis using the metamodels can be an alternative to reduce the effective input dimension by jointly selecting influential variables and the approximation model [34].

2.2.4 Impact of Control on System Architecture or Component Design

A key aspect of power-electronics design is the strong coupling between control strategy and hardware performance. A topology or component configuration cannot be meaningfully optimized in isolation, as its efficiency, loss distribution, thermal behavior, and EMC characteristics depend directly on the applied control. Ignoring control therefore leads to biased or incomplete design conclusions.

Control should be treated as an integral dimension of the design space rather than a downstream step. Advanced control strategies can reshape system behavior and compensate for hardware limitations. For example, an intelligent controller can shift the balance between conduction and switching losses, allowing a topology with high conduction losses but favorable switching characteristics to operate in regimes where its weaknesses are mitigated and its strengths are exploited. Thus, control effectively defines the achievable performance envelope of a system.

A representative example of this control–hardware coupling can be found in modern automotive traction inverter design, where operating strategies directly influence system-level trade-offs. This interaction is evident in high modulation index operation in automotive traction inverters, particularly six-step or overmodulation modes [35, 36]. Extending operation beyond the linear

range increases the fundamental output voltage, enabling higher power or reduced current, with system-level impacts of approximately 10%. At the same time, it alters loss distribution, harmonic content, and thermal stress across components.

Control consideration is not a binary choice but a continuous, highly parameterized space, as seen in the pulse pattern example [37]. This expands the optimization problem to a coupled hardware–control space, making traditional approaches insufficient. Consequently, AI-based design must inherently *co-optimize control and hardware*. In particular, for topology exploration, performance can only be assessed under a suitable control strategy. Integrating control synthesis into the design loop is therefore essential to ensure meaningful, closed-loop evaluation of candidate architectures.

2.3 Multi-Physics & AI-Enhanced Co-Design and Optimization (Component Level)

2.3.1 Semiconductor, Magnetics, and Passive Surrogates

At the semiconductor-device level, design already involves strongly coupled trade-offs among multiple performance metrics and device parameters, making efficient multi-objective exploration increasingly important [38]. Similar demands arise for magnetics and passives, where accurate but computationally efficient models are needed to integrate component behavior into broader multi-physics and system-level optimization [38]. Magnetic components – inductors, transformers, and EMC filters – are key determinants of efficiency, power density, and thermal behavior in modern power-electronics systems. Especially in WBG-based converters with high switching frequencies, fast edges, and multi-tone excitations, accurate yet fast core-loss models are essential for component-level co-design with semiconductors, cooling, and mechanics. This creates a need for surrogate models that retain physical insight while being computationally cheap enough for large-scale optimization loops. Classical core-loss models, such as Steinmetz-type formulations and their generalized variants, approximate the specific loss as a function of frequency and peak flux density and can be extended to non-sinusoidal waveforms. They are compact, interpretable, and straightforward to parameterize for individual materials. However, they struggle once arbitrary pulse-width modulation (PWM) patterns, multi-frequency excitation, DC bias, temperature dependence, and cross-material generalization are required (even though some modified formulas have been presented in academia). Hysteresis and finite-element-based models improve fidelity but are too computationally expensive for iterative, system-level design-space exploration.

The HARDCORE approach in [39] exemplifies how hybrid, physics-informed machine learning can bridge this gap for toroidal ferrite cores under arbitrary $B(t)$ waveforms. The problem is decomposed into two stages. First, a 1D residual, dilated convolutional neural network reconstructs the full time-series of the magnetic field $H(t)$ from the input flux density $B(t)$ and engineered features. Second, the power loss is obtained by physical integration of the B–H loop area, followed by a small residual correction network to compensate remaining systematic errors. A central methodological element is extensive feature engineering: in addition to the raw $B(t)$ sequence, the model ingests derivatives, waveform classifications, and physically meaningful scalar descriptors such as frequency and temperature. These are fused with CNN-extracted temporal features in a lightweight architecture, enabling high prediction accuracy with very few parameters and modest data requirements.

Embedded in a multi-physics co-design workflow, such magnetic surrogates provide fast, physically grounded loss estimates across wide design spaces (material, geometry, gap configuration, drive conditions). Coupled to thermal and reliability models, they enable efficient evaluation of hotspot

temperatures and lifetime within automated, AI-driven optimization and topology exploration, turning magnetics into a first-class design knob rather than a post hoc constraint.

2.3.2 Hybrid ML–Physics Solvers for Coupled Electrical–Thermal–Mechanical Design

Beyond the component-level surrogates discussed above, co-design of power-electronics modules requires evaluating tightly coupled multi-domain behavior. Electrical losses produce spatially non-uniform heating; the resulting thermal gradients drive mechanical stress because different material layers – semiconductor die, solder, copper traces, ceramic substrate – expand at different rates. This coupled electrical–thermal–mechanical (E–T–M) interaction must be resolved during design, and full-fidelity FEA or CFD remains the standard tool for doing so. However, a single parametric design sweep can demand thousands of solver evaluations, making brute-force exploration impractical when multiple objectives must be balanced simultaneously.

Classical model order reduction (MOR) has long been used to ease this bottleneck. Projection-based methods compress the full finite element system onto a small mathematical basis, reducing tens of thousands of equations to a handful while preserving the solver's mathematical structure and error bounds. For SiC power modules, parametric MOR applied to directly coupled thermal–mechanical FEA – where temperature and deformation are solved simultaneously – has demonstrated computation-time reductions of roughly 80 % at sub-percent stress error, enabling rapid design-point sweeps over material properties and layer geometries [40]. These methods are well-established and trustworthy but remain limited: they require expert setup, and their efficiency degrades for strongly nonlinear or high-dimensional parameter spaces.

Machine learning is beginning to enter this workflow at several levels. The most mature route uses the physics solver to generate training data and fits a data-driven surrogate – typically a neural network, Gaussian process, or polynomial expansion – that replaces the solver in subsequent optimization loops. For IGBT heat sink design, ensembles of polynomial chaos expansion surrogates trained on three-dimensional CFD have enabled geometry optimization with built-in uncertainty quantification [41]. For converter-level parameter design, a recent framework chains a neural-network feasibility classifier with a probabilistic efficiency-and-temperature predictor and a multi-objective optimizer, all trained on simulation outputs, to navigate efficiency–temperature trade-offs under thermal constraints [42]. In both cases the physics solver remains the source of ground truth and is re-invoked to validate final candidates; the ML model accelerates the search but does not replace the physics. This distinguishes simulation-trained surrogates from the physics-informed neural networks discussed later in this chapter, where the governing equations themselves are encoded in the training process.

What remains largely missing, however, is the deeper integration that the term "hybrid ML–physics solver" implies: architectures where a machine-learning model and a physics solver operate jointly within a single solution process – for instance, ML-corrected reduced-order models, neural-operator-augmented time steppers, or learned preconditioners that accelerate iterative FEA solvers. Such approaches are advancing rapidly in adjacent fields such as structural mechanics and additive manufacturing process simulation, but their application to coupled E–T–M problems in power-electronics module design is still at an early stage. A multilevel multiphysics framework on the open-source MOOSE platform has demonstrated coupled electrical–thermal–mechanical simulation with reduced-order modeling for multichip electronics co-design, identifying ML augmentation as a natural next step [43].

The practical incentive for closing this gap is clear. Evolutionary or genetic-algorithm-based optimization of complex cooling structures – such as additively manufactured heat sinks with topology-optimized internal geometries – typically requires hundreds of sequential CFD evaluations, with total computation times measured in days to weeks [44]. Embedding genuinely hybrid ML–physics solvers into the inner evaluation loop of such optimizers could compress these campaigns substantially, making iterative, application-specific E–T–M co-design viable within realistic project timelines. This acceleration is a key enabler for the multi-objective and evolutionary optimization strategies discussed in the following section.

2.3.3 Evolutionary Optimization of Components, EMC Filters, and Layout

The optimization campaigns just described illustrate a broader pattern: power-electronics design problems are inherently multi-objective, and the relevant objectives almost always conflict. A heatsink must balance thermal resistance against pressure drop, mass, and temperature uniformity. A power module layout must trade parasitic inductance against thermal performance and current sharing. An EMC filter must satisfy emission limits while minimizing volume, loss, and cost [45]. No single scalar merit function captures these trade-offs, which is why Pareto-based optimizers have become the standard framework, with evolutionary algorithms such as genetic algorithms (GA) and non-dominated sorting genetic algorithm (NSGA)-II as the most widely used engines.

In practice, these algorithms are coupled directly with physics solvers so that every candidate is evaluated at full fidelity. The approach spans a broad range of components. GA-coupled CFD and NSGA-II with analytical thermal models have been applied to heat sink and radiator geometry optimization [44, 46]. Evolutionary search paired with two-dimensional FEA has automated planar transformer design targeting hard-to-optimize parasitic elements [47]. Comprehensive EMC filter platforms sweep component databases against high-frequency parasitic models to evaluate topologies for volume, loss, and attenuation simultaneously [48]. At the module level, NSGA-II has been used to position bare dies and wire bonds within multi-chip layouts, jointly minimizing stray inductance, branch-inductance mismatch, and junction temperature spread with automated electromagnetic and thermal simulations [49]. Design-automation tools such as PowerSynth 2 extend this paradigm with constraint-graph-based layout generation and reduced-order models, enabling rapid synthesis of 2D through 3D module layouts [50].

Several gaps limit broader adoption. Evaluation cost remains the dominant bottleneck: thousands of solver calls at minutes to hours each restrict the design variables and objectives that can be explored within project timelines. Bayesian optimization – a sample-efficient alternative well established in adjacent domains – remains largely unexplored for power-electronics components. Equally absent is the systematic multi-objective sizing of passive components under coupled electrical, thermal, and volumetric constraints. The faster surrogate and physics-informed models discussed in the following section could alleviate the evaluation-cost bottleneck and make both evolutionary and Bayesian campaigns viable at realistic converter complexity.

2.4 AI-Accelerated Simulation, Reliability Prediction, Design Validation, and Verification

2.4.1 Surrogate and Physics-Informed Models

Power-electronics design increasingly requires large numbers of high-fidelity electromagnetic and thermal simulations for design-space exploration, MOO, tolerance and Monte Carlo studies, and geometry or topology optimization. Classical finite-element and multi-physics solvers provide trustworthy results, but they become a bottleneck when thousands to millions of evaluations are required. Similarly, although SPICE-level circuit simulations are typically efficient for a limited number of runs, they can also become time-consuming in large EMC-oriented parameter sweeps or optimization loops. At the same time, engineers increasingly need to include measured nonlinear material and device behavior in their models, which is cumbersome in purely equation-based solvers but natural for machine-learning surrogates [51]. Fast, data-aware surrogate and physics-informed models therefore become key enablers for simulation, tuning, and virtual qualification, especially as scalar design parameters give way to geometric and even topological degrees of freedom.

PINNs [26] and related methods, such as physics-informed DeepONets and neural operators [52], approximate the solution of partial differential equations directly through a neural network rather than repeatedly solving the discretized system. During the initial, one-time training phase, the network parameters are optimized to minimize a composite loss that enforces the governing-equation residuals, as well as deviation from boundary or initial conditions [26]. Automatic differentiation provides efficient derivatives, and physical constraints can be hard-constrained into the architecture [53]. Once trained, evaluation is reduced to a forward pass, avoiding an iterative solver and enabling inference times in the range of 1–100 ms for full 2D or 3D fields in reported examples [53, 54, 55, 56]. The main benefits are direct coupling of multiple physical domains, ultra-fast inference suitable for optimization loops, and native inclusion of measurement data [26, 53]. Remaining challenges include robust training for industrial problems, appropriate scaling and sampling, architecture selection, optimization strategy, and the treatment of sharp interfaces and stiff physics [53].

Recent applications demonstrate both the potential and the current limits of physics-informed surrogates for multi-physics problems relevant to power electronics:

- *Magnetostatic transformer surrogate for inductance and loss:* For axisymmetric transformer geometries, convolutional PINNs approximate the magnetic vector potential over an 18-dimensional geometric parameter space, including different coil and ferrite shapes [54]. Reported inference times are 8–30 ms per geometry compared with approximately 0.9 s for the finite-element reference, corresponding to a speed-up of about 30–100× while maintaining less than 1% deviation in inductances and coupling factors [54]. Coupled with a data-driven loss model such as HARD-CORE, core losses can be predicted in less than 50 ms per sample [39]. These results indicate that power-electronics-relevant field quantities, such as inductance, coupling, and core losses, can be evaluated at optimization timescales.

- *Electrochemical battery model for mission-profile simulation:* A physics-informed DeepONet surrogate for a single-particle lithium-ion battery model was trained in a purely physics-informed, data-free fashion to capture the voltage response under arbitrary current profiles and eight scalar electrochemical parameters. Training used 10,000 randomly generated current profiles, 50,000 collocation points in a 10-dimensional spatio-temporal-parameter domain, 1,000,000 epochs, and approximately 80 h on an NVIDIA H200 GPU. The trained model yielded mean voltage errors

below 1 mV for constant, pulse, and drive-cycle profiles, with CPU inference around 3.5 ms, about 100× faster than the numerical reference solver [55].

- *Multi-domain integration and optimization with PINN-based surrogates:* The above examples generalize to multi-domain power-electronics workflows, where electrical, electromagnetic, and thermal models must be combined under mission profiles and design variability. Physics-informed neural operators allow training of parametric solution operators over geometry, material, and operating conditions [53], and they can be embedded into higher-level tasks such as inverse design and topology optimization [56], design of experiments and parameter estimation [55], and transformer optimization across an 18-dimensional geometric space [54]. Existing results show good in-distribution accuracy and useful, but not unlimited, extrapolation. Large topological changes or strongly different physics can degrade model quality. Thus, PINN-based surrogates are already strong candidates for inner loops of MOO and mission-profile-based reliability analysis, but improved training strategies, robust handling of sharp interfaces and non-smooth phenomena, and systematic uncertainty quantification are still required before they can replace finite-element models in safety-critical decision loops.

Physics-informed surrogates are progressing from research prototypes toward deployable components in industrial workflows. Existing frameworks such as DeepXDE, NVIDIA Modulus, and TorchPhysics already provide physics-informed neural-network building blocks with GPU support and advanced training strategies. Ongoing research on data-calibrated neural networks for semiconductor and power-electronics manufacturing, as well as deployment on edge devices for automation or control, points toward integrated and certifiable surrogate modules embedded in virtual design environments such as electronic design automation (EDA) and technology computer-aided design (TCAD). For power-electronics simulation and reliability prediction, this evolution promises standardized libraries of validated surrogates for EMI, magnetics, and thermal behavior, enabling real-time virtual prototyping, automated design exploration, and more robust lifetime assessment.

2.4.2 Virtual Twins and/or Shadows

Virtual twins (also known as digital twins) extend classical simulation models by linking design data, operating conditions, measurements, and reliability models into an updatable representation of the power-electronics system. In the context of this section, their main role is to connect fast surrogate and physics-informed models with reliability prediction, lifetime assessment, and design verification. This is particularly relevant because the lifetime of a power-electronics system is rarely governed by a single component or stress factor. Semiconductors, bond wires, solder or sintered joints, substrates, capacitors, magnetic components, and cooling interfaces are exposed to coupled electrical, thermal, mechanical, and environmental loads [57, 58].

In a reliability-oriented virtual twin, converter operation and mission profiles are translated into losses, currents, voltages, temperatures, and stress cycles. Thermal impedance networks, field simulations, or surrogate thermal models then provide component-level temperature profiles. These profiles are coupled with degradation or lifetime models to estimate damage accumulation, remaining useful lifetime, or probability of failure. For power modules, dominant failure mechanisms are often linked to junction-temperature swings, mean temperature, power cycling, and thermo-mechanical fatigue in interconnects and packaging. For passive components, lifetime is typically governed by hot-spot temperature, ripple current, voltage stress, humidity, and material aging.

This makes the virtual twin useful across several stages of the design process. During concept and layout selection, it can compare candidate topologies, cooling concepts, packaging technologies, and control strategies under the same mission profile. During verification, it can check whether expected operating stresses remain within design limits. During application, it can be updated with measurement data to detect deviations between predicted and observed behavior. AI contributes mainly by accelerating expensive model evaluations, updating model parameters from measured data, detecting anomalies, and estimating uncertainty. However, the virtual twin should not be treated as a single black-box lifetime predictor. For engineering use, it must preserve traceability from requirements and design parameters to simulated stress, validation data, lifetime models, and final reliability decisions.

2.4.3 Verification / Validation of the Delivered Results

AI-accelerated simulation, surrogate models, and virtual twins can support industrial design only if their results are trustworthy, reproducible, and traceable. Verification and validation address different questions. Verification asks whether the model implementation, software workflow, data pipeline, and numerical procedures are correct. Validation asks whether the delivered result is sufficiently accurate for the intended engineering decision. For AI-based tools, this distinction is essential because the output depends not only on physical equations and solver settings, but also on training data, model architecture, hyperparameters, uncertainty treatment, and extrapolation behavior.

A practical verification and validation workflow should start with a clear definition of intended use. A surrogate used for early design-space screening does not require the same evidence level as a model used for lifetime prediction, compliance support, or safety-relevant decision-making. The validation envelope should therefore specify the allowed ranges of voltage, current, temperature, switching frequency, topology, component technology, cooling condition, mission profile, and layout assumptions. Predictions outside this envelope should be explicitly flagged as extrapolation rather than reported with the same confidence as validated in-domain results.

Automated verification and validation can reduce manual effort by embedding consistency checks directly into the design workflow. These checks may include unit and boundary-condition checks, comparison against analytical limits, conservation-law residuals, monotonicity or passivity constraints, independent solver cross-checks, regression tests across software versions, and comparison with measurement data. For AI models, additional checks are required, including uncertainty quantification, calibration of confidence intervals, sensitivity analysis, explainability of dominant input factors, out-of-distribution detection, and documentation of training and validation datasets. Current AI quality and risk-management standards increasingly emphasize such lifecycle-based assessment, including quality characteristics, risk identification, monitoring, and treatment of AI-specific uncertainty [59, 60].

Compliance must also be addressed explicitly. Depending on the application, power-electronics products may be subject to IEC, ISO, UL, automotive, grid-code, or customer-specific requirements. AI-generated results should therefore be linked to requirements, test cases, model assumptions, and evidence artifacts. For safety-relevant functions, IEC 61508 emphasizes structured lifecycle processes for electrical, electronic, and programmable-electronic safety-related systems; AI-based tools must be integrated into such processes rather than used as untraceable shortcuts [61]. This does not imply that every AI surrogate must itself be certified, but its role must be clearly classified, for example as design assistance, screening, virtual testing, qualification support, or control and protection functionality.

Virtual qualification and AI-generated stress scenarios are promising extensions of this workflow. Instead of only reproducing known test cases, AI can help generate corner cases, rare mission profiles, worst-case thermal cycles, tolerance combinations, EMI-sensitive operating points, and aging-relevant stress patterns. These scenarios can improve design robustness and reduce the risk that validation only covers nominal operation. Nevertheless, virtual qualification cannot replace physical testing unless the correlation between model prediction and measured behavior has been demonstrated. The credible route is therefore hybrid: AI accelerates scenario generation, simulation, and documentation, while final qualification remains anchored in independent measurements, physical constraints, uncertainty bounds, and auditable engineering judgment.

2.5 Outlook and Research Needs

2.5.1 Summary – Integrated Design–Simulation–Qualification Digital Twins

The previous sections indicate a gradual shift from isolated AI tools toward integrated design–simulation–qualification environments. In such a workflow, topology exploration, component sizing, multi-physics simulation, reliability prediction, and verification are no longer treated as separate sequential steps. Instead, they are connected through a digital twin framework in which design parameters, simulation models, surrogate models, mission profiles, measurement data, and qualification evidence remain traceable throughout the development cycle.

For power electronics, this integration is particularly important because design decisions are strongly coupled across domains. A topology or modulation strategy affects semiconductor losses, thermal loading, magnetic design, EMC behavior, capacitor stress, and ultimately lifetime. Integrated twins can therefore provide a common evaluation layer: AI-assisted topology or parameter candidates are first screened by fast surrogate and physics-informed models, then assessed under mission profiles, and finally linked to reliability and verification criteria. This enables earlier identification of weak design regions and reduces the risk that optimization improves one metric while violating thermal, EMI, lifetime, or manufacturability constraints.

The long-term objective is not a fully autonomous black-box design flow, but a traceable and certifiable engineering workflow. Digital twins should combine physics-based models, data-driven surrogates, uncertainty quantification, and measured validation data. Their output should support design decisions, generate evidence for virtual qualification, and guide physical testing, rather than replacing engineering judgment. In this role, AI becomes the acceleration and coordination layer: it reduces simulation effort, proposes design alternatives, identifies stress scenarios, supports documentation, and helps maintain consistency between requirements, models, tests, and qualification artifacts.

Consequently, future power-electronics design environments will likely evolve toward closed-loop digital engineering platforms. These platforms will combine automated exploration, fast multi-physics evaluation, reliability-aware optimization, and standards-oriented verification. Their industrial value will depend less on individual AI model accuracy alone and more on traceability, robustness, uncertainty handling, and integration into existing engineering and certification processes.

2.5.2 Impact of LLMs on the Design Processes

The application of LLMs and transformer-based foundation models is increasingly reshaping engineering workflows. These models are built on the transformer architecture introduced in Attention Is All You Need [62], whose self-attention mechanism enables scalable contextual reasoning and underpins modern generative pre-trained transformer (GPT)-class systems [13]. Large-scale pre-training and transfer learning have become dominant paradigms, while recent multimodal models extend these capabilities to cross-domain inputs relevant to engineering, including textual, graphical, and simulation data [63, 64].

In power electronics, the adoption of LLMs shows a clear asymmetry between software-centric and hardware-centric workflows. LLMs are already highly effective in software-oriented tasks such as requirement capture, embedded firmware generation, debugging, test synthesis, and documentation, benefiting from large code corpora and mature DevOps tooling [21, 65]. In contrast, hardware-centric activities, such as topology selection, component sizing, and simulation-driven optimization, remain less mature, with current demonstrations largely limited to research environments (e.g. PE-GPT and related frameworks) [28, 66]. This gap reflects the fundamentally higher complexity of hardware design, which requires integration of physical constraints, structured engineering knowledge, and tight coupling with simulation tools [21].

A key emerging direction is the use of autonomous and agent-based AI systems to orchestrate design workflows. Building on LLM capabilities for code generation (e.g. Codex-like systems), these agents can plan design tasks, generate simulation scripts, execute them in tools such as LTspice, SIMetrix, PLECS, or MATLAB/Simulink, analyze outputs, and iteratively refine design parameters. Recent approaches combine reasoning and acting (e.g. ReAct-style frameworks) with reflective optimization (e.g. Reflexion), enabling closed-loop design exploration across multiple tools and domains [67, 68]. Early implementations of personal AI assistants and engineering copilots (e.g. OpenClaw-type systems) demonstrate how such agents can automate multi-step workflows, including model setup, parameter sweeps, sensitivity analysis, and documentation generation. While still nascent, these agentic approaches represent a significant shift from passive assistance to active design orchestration.

Despite this progress, several limitations constrain adoption in hardware-centric workflows. Even physics-informed LLM agents require high-quality domain-specific data, formal validation procedures, and human oversight to ensure correctness and reproducibility [69], [70]. Approaches such as PE-GPT integrate retrieval-augmented generation (RAG), physics-based constraints, and simulation feedback loops to reduce hallucinations and enforce physically consistent outputs [21, 28]. Broader reviews highlight ongoing challenges in data availability, generalisation, and integration with existing engineering toolchains, necessitating systematic verification, corner-case testing, and expert validation [66, 71].

Consequently, LLMs are currently most effective when deployed as assistive systems within hybrid workflows. In such settings, LLMs accelerate ideation, documentation, and code-level development, while core hardware design remains grounded in physics-based modeling, simulation, and empirical validation, with engineers maintaining responsibility for verification and sign-off [21, 66]. Domain-adapted models (e.g. PE-GPT) illustrate how converter-specific constraints, modulation strategies, and design rules can be embedded into model reasoning, enabling more meaningful interaction with engineering tasks [28, 72]. When combined with digital twins and surrogate models, LLMs can support system-level reasoning consistent with electrical, thermal, and electromagnetic behavior [72].

Emerging capabilities further extend towards traceability and certifiable design workflows. LLM-based systems can map requirements to design artifacts, generate test cases, identify compliance gaps, and support documentation aligned with engineering standards [21], [65]. In parallel, multi-agent architectures enable distributed optimization across design tasks such as topology exploration, parameter tuning, and sensitivity analysis, providing a structured framework for coordinated design-space exploration [70, 67]. For industrial adoption, however, robustness remains critical: workflows must incorporate uncertainty quantification, explainability, and alignment with established IEC/ISO and automotive qualification processes [72]. Human-in-the-loop validation remains essential, particularly for manufacturability, reliability, and safety-critical assessment.

Looking ahead, the convergence of multimodal LLMs, agentic orchestration, simulation environments, and structured knowledge frameworks is expected to redefine design processes in power electronics. Future directions include automated modulation design, intelligent bill-of-materials optimization, EMI-aware layout generation, and physics-constrained topology discovery [69, 15]. As datasets, validation pipelines, and domain adaptation mature, LLM-driven systems – particularly those employing autonomous agents interacting directly with engineering tools – are likely to become foundational components of design environments, enabling faster iteration, improved traceability, and more globally optimized solutions [21, 72].

2.5.3 Call for Action: Skill Gap and Research & Industrial Needs

A persistent skill gap is emerging at the intersection of power electronics, data-driven modeling, and software-centric engineering. Beyond classical expertise in circuits and control, effective AI adoption requires competencies in machine learning, data engineering, and toolchain integration. Targeted upskilling, interdisciplinary teams, and engineering-ready AI practices are therefore prerequisites for deployment. This gap also contributes to existing skepticism and distrust toward AI-based approaches. However, this does not justify either rejection or blind trust; instead, adoption should proceed through incremental integration, with each step rigorously verified and validated within established engineering workflows.

From an industry perspective, the priority is to transition from manual, hardware-centric prototypes to robust, certifiable virtual workflows. This requires standardized data pipelines, validated datasets, and tight integration with existing EDA/computer-aided engineering (CAE) environments. Governance mechanisms – traceability, uncertainty quantification, explainability, and compliance with IEC/ISO/automotive standards – must be embedded by design. Organizations should invest in internal MLOps for engineering (data and version control, model validation, auditability) as well as in human-in-the-loop processes for verification and sign-off. In parallel, sustained upskilling programs and the progressive integration of AI into existing workflows are essential, with a strong emphasis on verification, robustness, and traceability at each step to systematically build trust in AI-assisted engineering.

From a research perspective, emphasis should shift toward reproducible, benchmarked, and system-level solutions. Critical gaps include the lack of public, high-quality datasets, limited cross-topology generalization, and insufficient integration of physical constraints into learning models. Priorities include hybrid physics–ML approaches, standardized benchmarks, open reference implementations, and scalable validation protocols that couple learning with simulation and experimental verification. Research institutes should act as multipliers, with the necessary courage to explore uncharted directions without being constrained by immediate economic justification. A strong focus on open, transparent, and reproducible models and pipelines is essential to enable rapid transfer

and adaptation in industry, reduce duplication of effort, and build collective trust in AI-driven engineering.

Funding and policy should be re-aligned toward infrastructure and translation: open datasets, shared simulation platforms, and reference toolchains; long-term programs that bridge AI, power electronics, and software engineering; and incentives for academia–industry collaboration. The focus should prioritize long-term, sustainable adoption of AI over short-term "quick wins" that rarely translate into industrial practice. Proactive programs addressing future challenges of AI – particularly generative AI – should aim to reduce ambiguity, improve robustness, and lower integration barriers. Support should explicitly target verification frameworks, certification pathways, and governance models that are practical for industry, including smaller entities, where excessive regulation combined with limited expertise can hinder adoption. Clear guidelines, shared best practices, and accessible validation infrastructures are essential to enable broad, trustworthy deployment of AI-assisted design.

Coordinated action across education, industry, and policy is required to *close the skill gap*, establish reliable engineering frameworks, and *enable trusted, scalable AI-driven design* in power electronics.

3 AI in Power Electronics Manufacturing

Growing demand for power electronics is driven by the energy transition and the electrification of transport, with appliances and consumer-device applications forming an additional major deployment area [73, 74]. This challenges manufacturers to realize smaller packages, higher integration density and improved thermal management while simultaneously increasing yield and lowering cost [75, 76, 77]. Production chains span tightly coupled stages from semiconductor fabrication and metrology to module, substrate, package integration and reliability testing [78, 79, 75]. Because the underlying data is multi-dimensional, nonlinear and interdependent, AI methods are increasingly used to support process understanding and optimization [80, 81, 78]. Reported benefits include faster yield ramp-up, reduced metrology delay, improved process quality and lower downtime [81, 82, 79, 83].

3.1 Systematization of AI Use-Cases and Methods in Power Electronics Manufacturing

To map where and how AI creates value in power electronics manufacturing, this chapter adopts the two-dimensional classification framework proposed by Chen et al. [84]. One axis captures the *cognitive dimension*, that is, the depth of analytical insight an AI system provides. The other describes the *structural dimension*, or the part of the manufacturing system to which that insight is applied.

3.1.1 The Cognitive Dimension: From Visibility to Autonomy

The cognitive axis follows the "four-knows" taxonomy, which classifies AI systems by the maturity of their analytical capability.

At the most basic level, *know-what* (*visibility*) establishes awareness of the current process state: whether equipment is drifting, whether a wafer remains within specification, or whether a process parameter has moved beyond its control limits. In power electronics manufacturing, where tolerances are narrow, and reliability requirements are stringent, this kind of real-time visibility already has substantial practical value [78].

Know-why (*transparency*) moves beyond detection and asks what caused the observed deviation. Instead of responding to symptoms through trial and error, root-cause models help engineers trace a defect back to its origin. This matters especially in multi-step process chains, where yield loss may be introduced several operations upstream of where it is finally detected [84].

Know-when (*prediction*) adds a temporal perspective. By forecasting future system states, whether product quality trends, equipment degradation trajectories, or process drift, manufacturers can shift from reactive responses to condition-based strategies. The operational benefit is clear, less unplanned downtime, better-timed maintenance, and more stable production output [79, 85].

The most advanced level, *know-how* (*prescription*), closes the loop. At this stage, AI systems recommend or autonomously execute corrective actions, from real-time recipe adjustments to optimized maintenance schedules. Although this promises the largest operational impact, deployment in power electronics remains limited by certification requirements and the demand for interpretability in safety-critical applications [84].

3.1.2 The Structural Dimension: Scope of Application

The second axis captures where these cognitive capabilities are deployed within the manufacturing system. At the *product level*, the focus is on die and module quality. AI supports yield improvement and reliability assurance by linking device characteristics such as electrical parameters, physical integrity, and thermal behavior to downstream field performance. The *process level* addresses individual manufacturing steps and correlates recipe parameters with step-level outcomes. This level of granularity is essential wherever small parameter shifts can have disproportionate effects on device reliability. At the *machine level*, attention shifts to the condition of the equipment itself by monitoring chamber health, consumable wear, and calibration drift to prevent unplanned downtime and preserve consistent process capability. The *system level*, finally, covers fab-wide concerns such as cross-step optimization, end-to-end traceability, and test-time reduction, all of which depend on integrating data across the manufacturing execution system (MES).

Overlaying both dimensions yields a matrix in which each cell represents a distinct class of AI application with its own data requirements, modelling challenges, and operational value. Table 3.1 presents this mapping for power electronics manufacturing. Because published AI applications targeted specifically at power semiconductor fabrication are still limited, the table also draws on validated work from adjacent semiconductor domains such as logic and memory manufacturing, where process equipment, inspection modalities, and data structures are largely shared.

Table 3.1: Representative AI/ML Applications in Power Electronics Manufacturing

Level	Scope	Know-what (visibility)	Know-why (transparency)	Know-when (prediction)	Know-how (prescription)
PRODUCT	***Front-End Testing***				
	Wafer Sort	Wafer defect classification [86, 87, 88]	Spatial pattern analysis [89, 90]	Virtual electrical test [79, 91]	Coverage-based test reduction [92]
	Back-End Testing				
	EoL Test	Parametric anomaly detection†	Failure mode clustering†	Reliability prediction†	Yield-optimised binning [93]
	Burn-In Test	In-situ degradation monitoring†	Failure mechanism classification†	Infant mortality prediction [94]	Burn-in reduction [95, 96, 97, 98]
PROCESS	***Front-End Processing***				
	Epitaxy	Growth state from in-situ signals [99]	Growth parameter correlation [100]	Epitaxial growth rate prediction [101]	Run-to-run recipe optimisation [102, 103]
	Ion Implant.	Beam shape control [104]	Dose uniformity root-cause†	Ion source remaining useful life (RUL) prediction†	Dose control [105]
	Annealing	In-situ sheet resistance monitoring†	Activation uniformity root-cause†	Heater RUL prediction†	Sheet resistance optimisation [106]
	Etching	optical emission spectroscopy (OES)-based endpoint detection [107]	Chamber fault classification [108]	Etch tool RUL prediction [109, 110, 111]	Etch recipe optimisation [112]
	Lithography	Wafer map defect detection [87]	Overlay error root-cause [113, 114]	Mask degradation prediction†	Dose/focus optimisation [105]
	Back-End Assembly				
	Die Attach	X-ray/scanning acoustic microscopy (SAM) void detection [115, 116]	Void formation root-cause†	Void-to-R_{th} prediction [117]	Pick-and-place optimisation [118, 119]

Continued on next page

Table 3.1 continued from previous page

Level	Scope	Know-what (Visibility)	Know-why (Transparency)	Know-when (Prediction)	Know-how (Prescription)
	Bonding	Bond lift-off monitoring [120]	Interpretable quality classification [121]	Pull/shear strength prediction [122, 123, 124]	Multi-objective bonding parameter optimisation [125]
	Encapsulation	Void detection [115]	Mold flow defect correlation [126]	Package warpage prediction [127]	Warpage parameter optimisation [128]
MACHINE	Equipment	Chamber state monitoring; tool drift detection [85, 129]	DL-based equipment fault diagnosis [130]	Downtime/RUL prediction [85, 131, 83]	Maintenance scheduling [84]
SYSTEM	Fab-Wide	Cross-tool fault detection and classification (FDC) transferability evaluation [132]	Cross-step root-cause analysis [80]	Yield forecasting [133]	Production scheduling [134, 135]

† Plausible use-case for which no published source could be identified at the time of writing.

3.2 From Insight to Implementation: Selecting the Right Model

Once a use-case has been identified, the next practical question is which model architecture to deploy. On the shop floor, that choice is rarely determined by accuracy alone. Data availability, latency constraints, and the need for interpretable outputs all carry substantial weight, particularly when model predictions inform process control or quality assurance decisions [84, 78, 121, 136]. Table 3.2 links common model classes to their typical data modalities and cognitive stages within the Four-Know framework [84], and serves as a reference for matching a given use-case to a suitable architecture.

Table 3.2: Representative AI Model Classes and Example Applications in Power Electronics Manufacturing (Compiled by the Authors Based on [84, 78, 79, 137, 103])

Model class	Methods	Cognitive stage	Data modality*	Example applications
Convolutional	CNNs, ViT, U-Net	Know-what	Images / Time-series	Wafer inspection; void detection; bond inspection
Autoencoders	AE, VAE	Know-what / Know-why	Images / Time-series	Chamber anomaly detection; drift identification
Tree-based	XGBoost, Random Forest	Know-why / Know-when	Tabular	Virtual metrology; quality prediction
Probabilistic	Bayesian networks, GPs	Know-why	Tabular	Parameter exploration; uncertainty quantification
Recurrent / Attention	LSTM, Transformers	Know-when	Time-series / Images	RUL prediction; thermal profile forecasting
Physics-informed	PINNs, hybrid ML	Know-why / Know-when	Time-series / Tabular	Thermal resistance estimation; digital twins
Policy-based	RL, Bayesian optim.	Know-how	Tabular / Time-series	Recipe tuning; run-to-run control
Clustering	k-means, SOM	Know-why	Tabular / Images	Defect clustering; equipment segmentation

* Most model classes can process other data modalities; listed entries reflect typical usage.

Across all four structural levels, production deployments already demonstrate the practical value of these model classes. Convolutional architectures now routinely reach 96–98% accuracy in wafer map defect classification [138, 139], and where labelled data is scarce, vision foundation models with few-shot adaptation have closed much of the remaining gap [140].

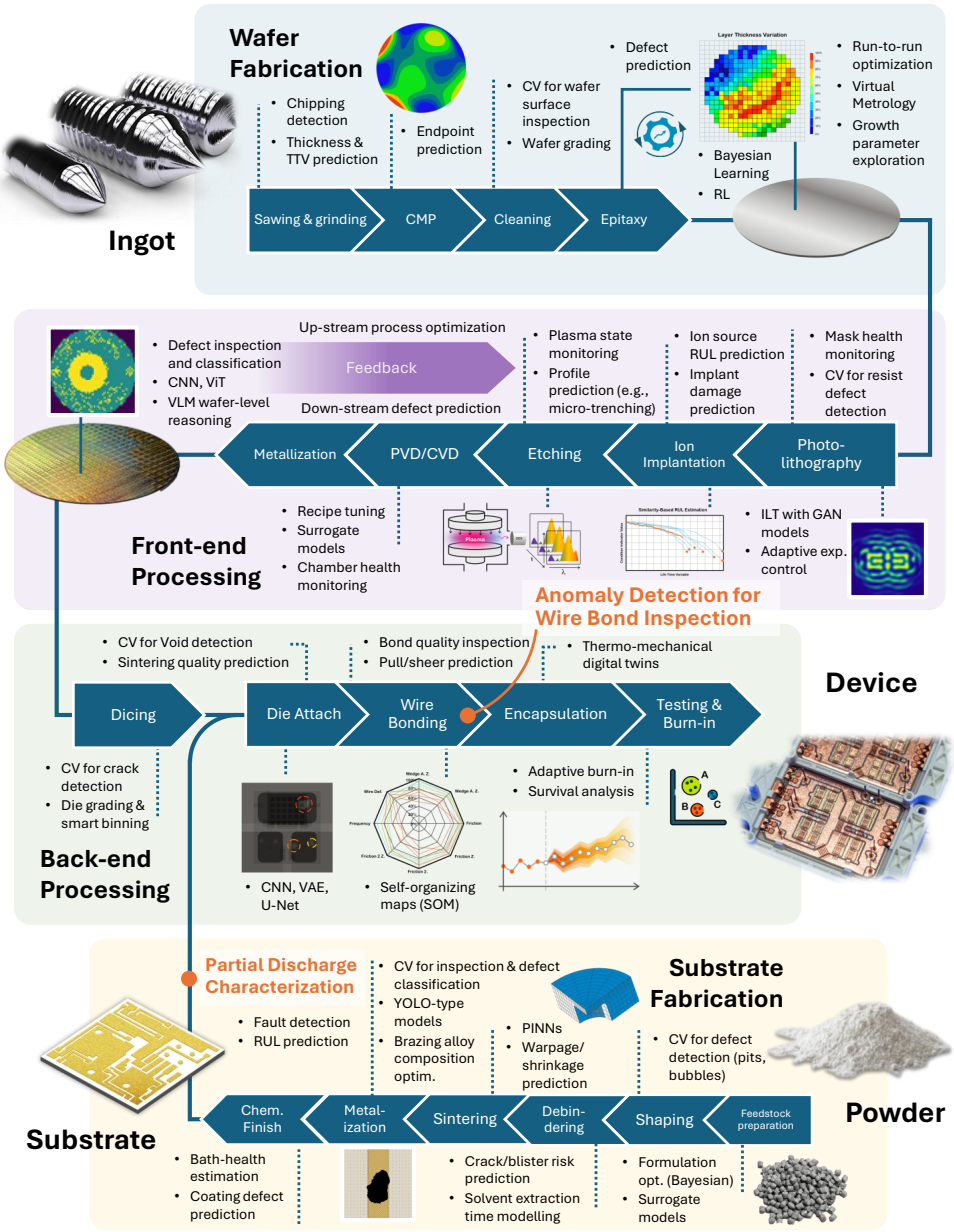


Figure 3.1: Overview of the power electronics manufacturing chain with representative AI/ML methods and applications mapped to each process step. *Image credits:* frog – stock.adobe.com (#47721432). Muhammad – stock.adobe.com (#2070511296). iStock.com/dahuang1231 (#2248314480).

Tree-based methods such as XGBoost dominate virtual metrology, enabling 100% wafer coverage and substantially tighter run-to-run control loops by replacing infrequent spot checks with per-wafer quality estimates [141, 142, 143, 144]. At the machine level, semiconductor prognostics increasingly rely on recurrent, transfer-learning, and attention-based models for remaining useful life prediction, particularly for ion-mill etching equipment [145, 109, 110, 111]. More broadly,

industrial predictive-maintenance deployments have been reported to reduce unplanned downtime by up to 50% and maintenance costs by about 40%; in semiconductor manufacturing, this matters especially because losses from unplanned downtime can reach millions of dollars per hour [146, 83]. At the prescriptive end of the spectrum, reinforcement learning is emerging as a viable alternative to classical EWMA-based run-to-run controllers for processes whose nonlinear dynamics are not well captured by conventional linear models [137]. Figure 3.1 brings the structural and cognitive perspectives together by mapping representative AI methods and applications onto the physical manufacturing chain, from substrate fabrication through front-end wafer processing, back-end module assembly, and final device testing.

3.3 Exemplary AI Applications and Best Practices for Successful Implementation

To complement the systematic overview presented above, this section illustrates the practical impact of AI in power electronics through two case studies. The first, from Fraunhofer IKTS, applies a Transformer-based architecture to predict substrate remaining useful life from high-rate partial discharge signals. The second, from the Siemens power module production line in Erlangen, deploys unsupervised anomaly detection for automated wire bond inspection. Together, they demonstrate how AI methods translate from research into measurable production gains.

3.3.1 AI-Driven Remaining Useful Life Prediction from Partial Discharge Data

Usage of image classification as a means of interpreting data from various PE characterization methods such as computed tomography (CT) or SAM is state-of-the-art and widely available. Thus, a different approach will be presented here to serve as an example of more specialized AI methods in PE.

As a PE module is being manufactured, the complexity of the system increases exponentially. This creates the need for failure prediction as early in the process as possible, before further potentially lost value is added. Since direct copper bonded (DCB) substrates are prone to defects caused by thermal cycling, leading to breakdowns in the field, characterizing the substrates can mitigate follow-up costs. CT and SAM provide a (time-consuming) solution but cannot characterize electrical properties which are central to module integrity. As an alternative, partial discharge (PD) measurement is being established.

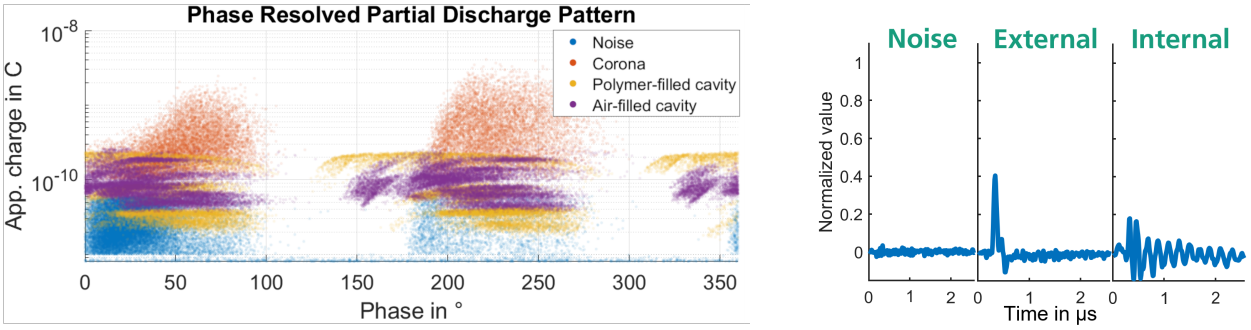


Figure 3.2: Phase-resolved partial discharge patterns, colored according to PD source (left) and characteristic pulse shapes (right).

PDs are discharges occurring in geometrically confined parts of an insulation system without bridging the whole insulation like a breakdown. They tend to occur at imperfections like voids,

cracks, conducting particles or material boundaries. When initiated by an AC source, each PD can be characterized by its intensity (called apparent charge q_{app}) and phasing in relation to the test voltage. Using these values, phase-resolved partial discharge (PRPD) patterns can be generated to form PD-source-dependent patterns. These are commonly used to identify the potential source of PD occurrence and are prone to overlapping of PDs from different sources, both in time and phase. To enable classification in these cases, the shape of each pulse can be used as it relates to the physical specifics of its occurrence (Figure 3.2).

The goal of the approach presented here was to find correlations between defects or RUL of DCB substrates and PD characteristics and train an artificial neural network (ANN) to detect degradation or defects. To provide the database, several hundred samples with double-sided metallization were produced and artificially aged batch-wise according to automotive qualification guideline (AQG) 324 from −40 °C to 125 °C at 15 min hold time. To classify the samples, an automated test stand enabling PD detection and pulse shape classification at up to 1.5 M events/s has been built at Fraunhofer IKTS. It enables testing conforming to IEC 60270 using user-defined voltage/frequency ramps reaching up to 17 kV at 20 Hz … 2 kHz.

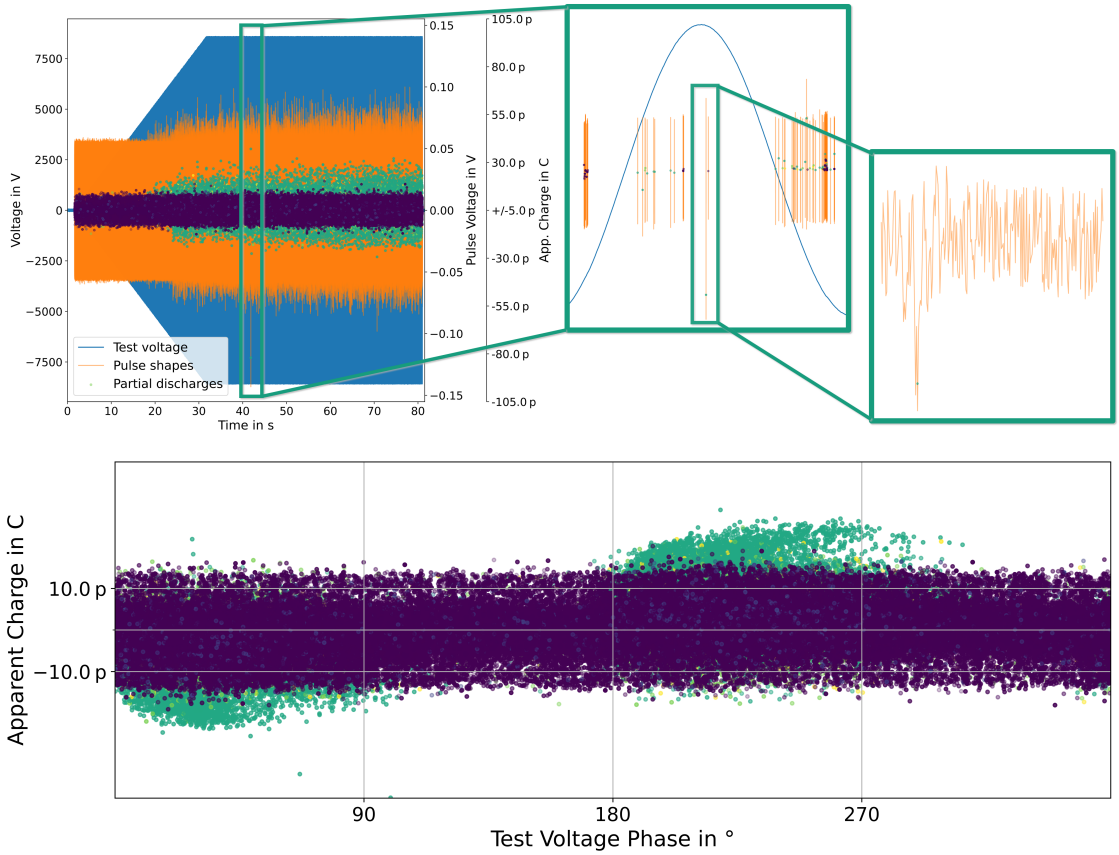


Figure 3.3: Various levels of detail in the time plot (top) and phase-resolved partial discharge pattern (bottom) of one sample which did not undergo artificial aging. Color-coding of PD events is done in real time using pattern classification.

Unlike conventional PD measurement systems, the solution developed at IKTS features sampling and real-time classification of the pulse shape acquired with each PD event. The resulting data is interdependent across multiple dimensions (Figure 3.3) and reaches data rates just below 800 Mbytes/s.

The gathered data is noisy, complex both in dimensionality and amount, overlaps, and has varying

information depth along the time axis which makes it somewhat similar to digitized human speech. Using this similarity, an artificial neural network used for both RUL and fault detection, was developed in cooperation with Impact Labs GmbH[1] within the project "Multimodal AI Analysis for Sustainable Power Electronics"[2]. The model uses a Transformer-based masked Autoencoder architecture which allows for resource-efficient outlier detection on whole pulse trains (multiple consecutive PD events). It is pre-trained on partially masked, unlabeled raw data from multiple measurement sources enabling it to learn the basic interdependencies of PD behavior and different discharge sources. The same model is then fine-tuned using labels from artificial aging or substrate defects which lowers the amount of manual labeling needed to fulfill regression or classification tasks. The system is pruned and quantized using methods established for speech recognition models to be able to run in real time on consumer-grade hardware instead of specialized AI clusters or hardware. It achieves more than 90 % RUL prediction precision (calculated based on thermal shock cycles) evaluating only 600 (!) pulse shapes from a measurement. ± 5 % (Figure 3.4).

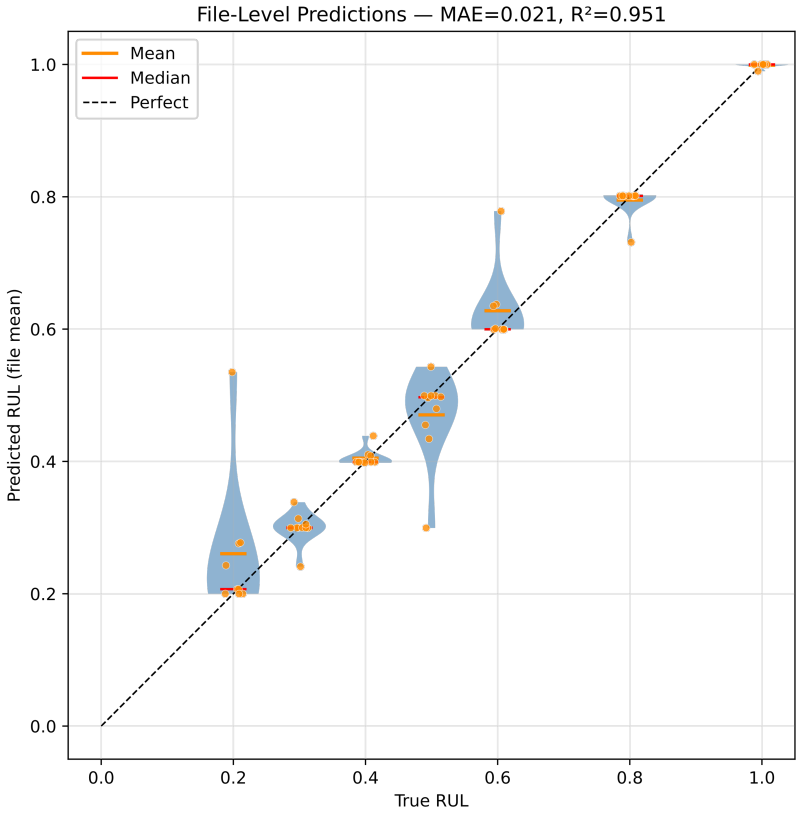


Figure 3.4: Predicted RUL compared to actual RUL of artificially aged SCB substrates. RUL is estimated linearly proportional to the amount of shock cycles.

The defect detection detects outliers based on fine tuning the model into recognizing the noise floor and golden board characteristics of the specific use case. This shows the high adaptability of the approach without the need to re-train the whole model which enables this approach to be adapted to raw measurement data from all kinds of sources.

[1] For further inquiries, contact Dr. Oliver Guhr at oliver.guhr@develappers.de

[2] Financed via Development Bank of Saxony and European Regional Development Fund under grant nr. 100723614

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Diese Maßnahme wird mitfinanziert durch Steuermittel auf der Grundlage des vom Sächsischen Landtag beschlossenen Haushaltes.

3.3.2 Unsupervised Anomaly Detection for Wire Bond Inspection in Series Production

Thick wire bonding represents a critical step in IGBT module assembly, where quality assurance has traditionally relied on 100% manual visual inspection. This labor-intensive process creates bottlenecks in production throughput and introduces variability due to operator fatigue and subjective assessment criteria. Typical defects—solder balls, fibers, misaligned bond wires, and contamination—can compromise module reliability if left undetected, yet their identification demands sustained attention across thousands of units per shift.

To address this, an automated optical inspection (AOI) system has been deployed in the power module back-end production line. Unlike conventional rule-based machine vision systems that require exhaustive programming of defect signatures, this implementation leverages unsupervised anomaly detection methods. The system operates without the need for defect-specific training data, instead learning a representation of "normal" bonding outcomes directly from production samples. Modules are automatically sorted into "normal" and "anomaly" categories based on deviation from this learned baseline, enabling immediate feedback to the bonding process.

The deployment has reduced manual inspection effort by more than 30%, shifting the operator role from exhaustive screening to spot-check validation and exception handling. Critically, the system generates anomaly score heatmaps that visually highlight the spatial location of detected deviations, enabling operators to rapidly identify and assess the specific nature of flagged defects during secondary review. This localization capability significantly accelerates root-cause analysis and supports more consistent decision-making compared to unaided visual inspection.

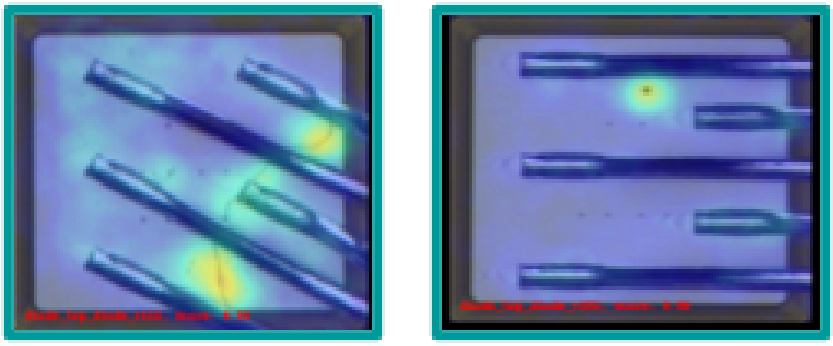


Figure 3.5: Anomaly score heatmaps of diodes with a fibre (left) and a solder ball (right). Courtesy of Nataliya Müller and Jonas Pirner, both Siemens AG.

From a cognitive perspective, the system establishes real-time visibility into process conformance at the product level, aligning with the *know-what* stage of the four-knows framework [84]. Its ability to generalize without defect-specific retraining addresses a persistent challenge in semiconductor manufacturing: the emergence of novel failure modes that were not anticipated during model development. By focusing on normality modelling rather than defect enumeration, the AOI system remains robust to process drift and unanticipated anomaly types [147].

The success of this application underscores a broader trend in AI-enabled quality assurance: the shift from supervised classification, which demands extensive labelled datasets, toward self-supervised and anomaly detection paradigms that can operate with minimal human annotation. Future extensions may incorporate *know-why* capabilities by linking detected anomalies to upstream process parameters such as bonding force, ultrasonic power, or substrate temperature, enabling predictive quality control [148, 121].

3.4 The Thinking Fab: Emerging Applications of Generative AI and Autonomous AI Agents

Generative and agentic AI represent a new phase in the evolution of artificial intelligence, characterized by systems that can create content and autonomously perform actions and pursue goals rather than only classify or predict. Prominent examples include image-generation systems like Stable-Diffusion, as well as text-generation models such as GPT and other LLMs that can write, summarize, reason, and generate complex documents or software code. More recently, these capabilities have been extended into agentic AI systems, which combine generative models with planning, memory, and tool-use mechanisms. Instead of producing a single output in response to a prompt, agentic systems can decompose complex goals into multiple steps, interact with software tools or databases, and iteratively refine their actions based on feedback. Applied to power semiconductor manufacturing and equipment maintenance, generative and agentic AI extend beyond traditional analytics toward adaptive, knowledge-driven production support and increasingly autonomous decision-making frameworks [149, 150, 151, 152].

Generative AI is beginning to address bottlenecks in both process development and operations, by replacing computationally expensive simulations with learned generative models, e.g., using image synthesis techniques to accelerate inverse lithography technology (ILT) workflows [153, 154, 155, 156] or generating high-quality synthetic data, to overcome data scarcity and improve the training of inspection and defect-classification systems [157, 158, 159, 160]. Building on these capabilities, agentic AI systems can orchestrate multi-step workflows across MESs, equipment control software, and data platforms, e.g., continuously monitoring tool health, correlating anomalies across process steps, triggering Predictive maintenance (PdM) actions, and coordinating responses such as recipe adjustments, or technician interventions [161, 151, 152, 162]. Together, these approaches enable a shift toward proactive, self-optimizing operations, where AI supports faster process development cycles, mitigates data limitations and assists engineers, extending traditional automation towards an adaptive and autonomous “thinking fab”.

3.4.1 The Transition from Analytical to Agentic AI

The practical implementation of AI systems ultimately culminates in operational decisions (e.g., pass/fail, go/no-go), either made by a human supported by AI-driven recommendations or conducted autonomously by the system itself. As the transition between different classes of AI used in such decision processes is gradual (since, at their core, most AI models estimate probabilities) we introduce the concept of “decision space” as a metric to distinguish between analytical AI, generative AI and agentic AI.

The decisions Space of traditional ML and Analytical AI, typically used in tasks such as defect classification, anomaly detection, or RUL prediction, is typically narrow and well-constrained. Decisions are discrete, often binary or categorical (e.g., pass/fail, defect type A/B/C, failure expected within X hours).

In contrast, generative AI operates within a significantly broader Decision Space. Instead of selecting from predefined classes, it produces high-dimensional outputs, such as images, signals, or text. For instance, in ILT applications, generative models define mask patterns at the pixel level, in synthetic data generation, they produce realistic wafer defect images. Language models generate code, recipes, or documentation token by token. Here, the decision space expands to continuous, structured output generation.

Agentic AI systems introduce a qualitatively different form of decision-making. By combining generative models with planning, memory and tool use, they operate in an open-ended decision space, which in principle is only bounded by available tools, system constraints, and imposed guardrails.

A striking illustration of both the potential and risks of such open-ended reasoning systems is OpenClaw, an experimental open-source agentic AI capable of interacting with diverse tools, executing shell commands, and operating with elevated system privileges [163]. In some reported cases, OpenClaw has autonomously resolved issues in its own codebase on a live system, demonstrating the ability to self-improve and handle unforeseen edge cases [164, 165]. However, there are also numerous reports of OpenClaw breaking its own code beyond recovery, highlighting the critical importance of robust guardrails, validation mechanisms and human-in-the-loop (HITL) control when deploying agentic AI in high-stakes industrial environments.

Table 3.3: Autonomy and Risk Categorization of AI Systems

AI category	Autonomy level	Decision space	Risk profile
Analytical	Low	Binary, discrete	Low (false-positive, false-negative)
Generative	Medium	Creative, informational	Moderate (hallucinations)
Agentic	High	Open-ended, functional	High (systemic instability)

3.4.2 Use-Case Categorization by Autonomy and Impact

To systematically structure AI applications, we map representative use cases onto a two-dimensional space defined by stakes/impact and agency/autonomy, as shown in Figure 3.6. The stakes/impact axis reflects the operational and financial consequences of a decision within the fab, while the agency/autonomy axis captures the extent to which an AI system can act beyond predefined logic, i.e., the breadth of its decision space. This framing compliments existing automation hierarchies (e.g., MES, advanced process control (APC), and equipment control) by explicitly linking AI capabilities to decision authority and risk.

At the low-agency end, AI systems operate within a constrained decision space, producing deterministic or categorical outputs. These systems are typically embedded in established fab workflows and tightly coupled to APC, FDC, and MESs. Representative high-impact use cases include defect inspection and classification, PdM with RUL estimation, and virtual metrology (VM). Although the decision logic is narrow (e.g., pass/fail, threshold violations, class labels), the operational implications are substantial: triggering a tool hold, rework loop, or maintenance action can directly affect yield, cycle time, and overall equipment effectiveness (OEE). As such, these Analytical AI systems occupy the high-stakes, low-agency quadrant, where robustness, interpretability, and validation are critical.

Conversely, high-agency systems, enabled by generative AI and ggentic AI architectures, operate over a significantly expanded decision space. These systems currently enter semiconductor manufacturing through the low- to medium-stakes applications, forming the "cognitive advisor" layer. Typical use cases include domain-specific LLM assistants for engineering support [166, 167], agentic RAG for document and knowledge search [168, 169, 161], design-of-experiments (DoE) support agents, as well as code and recipe generation aligned with equipment-specific syntax [170]. These systems integrate heterogeneous fab data, unstructured documentation, MES histories, FDC signals, and sensor time series, but remain primarily advisory, with decisions validated by engineers rather than directly executed on tools [171, 172]. This positioning allows rapid deployment and

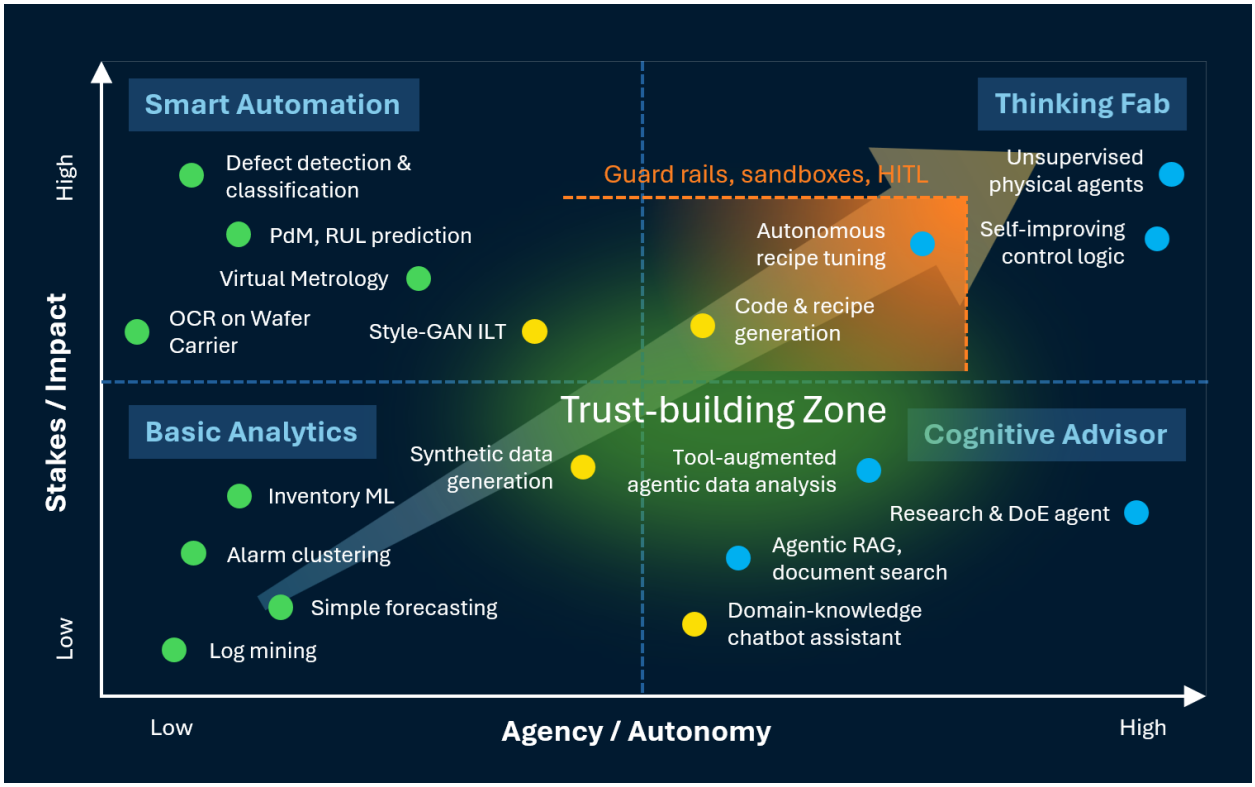


Figure 3.6: Use-case categorization by impact vs. autonomy. Green dots: analytical AI, yellow dots: generative AI, blue dots: agentic AI.

measurable productivity gains while limiting operational risk. In addition, generative AI enables synthetic data generation (e.g., wafer defect images, test data) [157, 158, 159, 160] and acceleration of computational lithography workflows (e.g., generative adversarial network (GAN)-based ILT approximations) [153, 154, 155, 156].

The high-agency, high-stakes quadrant represents the long-term trajectory toward the "thinking fab". Here, agentic systems evolve from advisory roles into closed-loop, autonomous control layers integrated with APC and equipment control systems. Emerging concepts include autonomous recipe optimization, self-adapting control strategies, and multi-agent coordination across process modules and toolsets [151, 173, 171].

Between these regimes lies a "trust-building-zone", where generative and agentic AI systems begin to influence operational decision without fully autonomous execution. Examples include tool-augmented agentic data-analysis, context-aware anomaly investigation, and early forms of recipe tuning support with APC frameworks [169, 150]. In this zone, autonomy is incrementally increased through the use of guardrails, sandboxed execution environments, and HITL validation. This staged deployment strategy aligns with semiconductor industry requirements for qualification, change control, and process stability, enabling gradual expansion of AI-driven authority.

As a practical example, Figure 3.7 illustrates the integration of agentic alarm data analysis in a software platform developed by InstruNEXT GmbH. The AI Agent operates with a local LLM and can use various tools such as statistics, correlation, anomaly and trend detection, and combine the results with context information retrieved from documents such as user manuals and component datasheets via agentic RAG, providing operators and engineers with actionable recommendations instead of just the alarm codes [174].

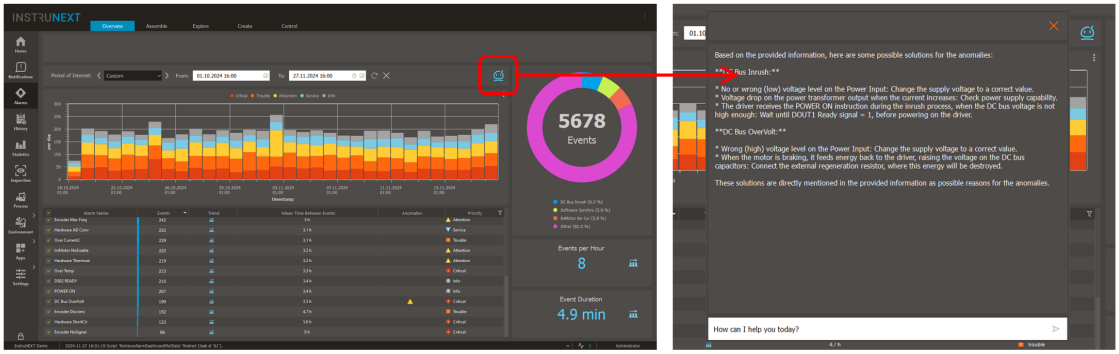


Figure 3.7: Agentic data analysis with tool usage and agentic RAG. Courtesy of InstruNEXT GmbH.

Ultimately, this may extend to unsupervised or minimally supervised physical agents capable of direct interaction with semiconductor equipment. While the potential benefits (self-optimizing processes, reduced variability, and improved yield) are significant, deployment in this quadrant requires stringent safeguards, including formal verification, runtime monitoring, and fail-safe mechanisms to meet the reliability standards of high-volume manufacturing.

In summary, AI adoption in semiconductor manufacturing follows a progressive expansion along the agency-impact spectrum: Analytical AI dominates high-stakes, low-agency applications within established MES/APC frameworks, while generative and agentic AI are introduced in the lower-risk advisory roles before advancing toward higher autonomy. This gradual enlargement of the Decision Space, supported by trust-building mechanisms and system integration, defines the pathway toward the *thinking fab*.

4 AI in Power Electronics Operations

This chapter addresses the use of AI methods and tools during product operation of power-electronic systems. In contrast to AI for design automation or offline analysis, operational AI must satisfy strict constraints in latency, determinism, robustness, safety, and maintainability. First, we explore typical tasks during online operation in section 4.1 and how AI methods can support those. Besides high-level method overviews and relevant literature references, we provide selected application examples. Then in section 4.2 we discuss industrial deployment challenges before providing a brief outlook into possible future AI-driven trends for power electronic system operation in section 4.3.

4.1 Typical Tasks during Product Operation

4.1.1 Control

AI methods in power-electronic control are mainly used to (i) improve dynamic performance under nonlinearities and constraints, (ii) reduce calibration effort across variants, and (iii) increase robustness against parameter drift. In product operation, these benefits must be realized under strict real-time constraints. Consequently, any AI component must be lightweight, deterministic in inference time, and bounded in its outputs [175]. Table 4.1 summarizes the main method classes, their typical role in the control architecture, and their current relevance for power-electronic product operation.

Table 4.1: Overview of AI-Assisted Control Methods for Power-Electronic Product Operation

Method class	Typical role in control	Representative PE use cases	Main advantage	Deployment readiness
Model-enhanced control [176]	Learn selected residual effects while retaining the classical control structure	Dead-time compensation, voltage-drop compensation, magnetic saturation compensation, disturbance estimation	Improved accuracy with preserved control structure	High
Learning-accelerated prediction / constraint handling [177]	Provide fast surrogate models or offline-learned policies for constrained control	Reduced-order prediction, approximate model predictive control, thermal- or efficiency-aware derating	Reduced online computation effort	Medium–High
AI-assisted tuning and scheduling [178]	Adapt controller parameters or operating modes across operating regions	Gain scheduling, adaptive parameter tuning, switching-frequency scheduling	Reduced calibration effort	High
RL-based supervisory control [179]	Select higher-level actions subject to validation by a rule-based supervisor	Setpoint scheduling, mode switching, efficiency-optimal operation	Long-horizon optimization and adaptation	Medium
End-to-end learned control [180, 181]	Directly map measurements to control actions	Primarily research-oriented demonstrations	Minimal explicit modeling effort	Low

Model-enhanced control: A practical entry point and industrially viable approach is to retain the classical control structure (current/voltage loops, phase-locked loop (PLL), modulator, protections) and introduce learning blocks only where modeling errors dominate. In this setting, learning is used as a targeted compensator rather than a full replacement. Typical examples are (a) data-driven disturbance observers (learning the lumped disturbance/uncertainty), (b) loss/thermal-aware derating maps learned from operation data, and (c) compensation of dead-time, voltage drops, and magnetic saturation via learned nonlinear functions. These approaches preserve a familiar safety envelope because the outer loop remains physics-based while the learning block is explicitly constrained (e.g., limited gain, limited bandwidth, monotonicity) [182], [183].

Learning-accelerated control and constraint handling: This is attractive in PE because key limits (current limit, dc-link ripple, thermal limit, switching states) can be handled explicitly. The main barrier, however, is computation under tight sampling deadlines. Two AI patterns are therefore common: (i) learning a reduced-order predictor (surrogate) for the plant or selected states to speed up prediction, and (ii) learning the control policy offline to replace online optimization with fast inference. In both cases, constraints should still be enforced explicitly (projection, clipping, admissible-set checks) rather than implicitly relying on the learned model [184], [185].

RL for tuning and scheduling: RL is most realistic when used for *supervisory* decisions rather than the inner current loop, since supervisory layers operate at slower rates and can tolerate additional checks. Examples include setpoint scheduling, efficiency-optimal switching-frequency selection, and adaptive gain scheduling across operating regions. A safe deployment pattern is that RL proposes an action, a rule-based supervisor validates it against constraints and stability margins, and fallback logic reverts to a baseline controller under uncertainty or abnormal conditions.

Stability, safety, and verification aspects: Because learning components are data-driven, validation must explicitly cover edge cases such as operating limits, fast transients, sensor dropouts, and saturation events [186], [187]. A representative application example is illustrated in Figure 4.1, where AI acts as a supervisory layer in inverter control by processing measured signals for fault detection and diagnosis. In this role, the AI block supports functions such as fault ride-through and transition to islanded operation when abnormal conditions are detected in grid-connected systems, while the inner control loops remain physics-based and deterministic. This example reflects a particularly relevant deployment pattern for product operation because it improves situational awareness and operational adaptability without transferring hard real-time actuation authority to the learning component. For control-relevant AI blocks, recommended engineering measures are:

- bounded outputs and bounded rate-of-change (to prevent discontinuities injected into the loop),
- input validity checks (range, plausibility, signal-quality flags),
- uncertainty-aware operation (increase safety margins when inputs are out-of-distribution),
- structured fallback to classical control (fail-operational or fail-safe depending on application).

Overall, the most deployable AI approach in control is a *hybrid approach*, where physics-based control provides the stability structure, and learning improves modeling and adaptation where the physical model is incomplete.

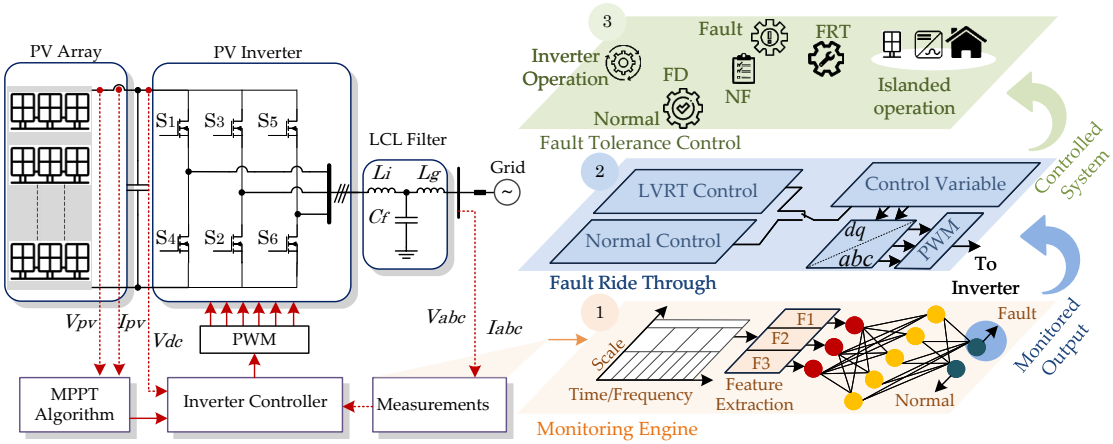


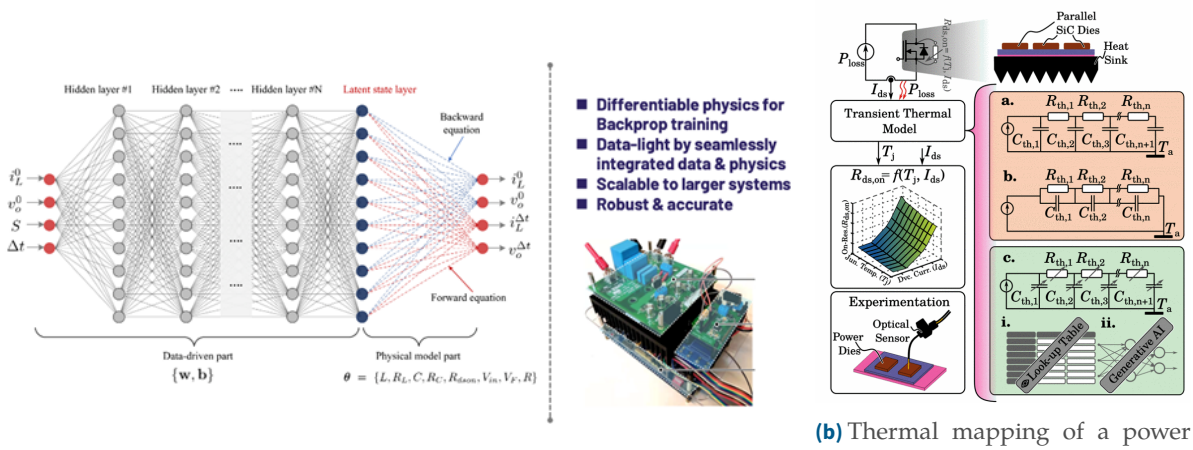
Figure 4.1: Representative example of AI-assisted supervisory inverter control in a grid-connected system. The AI layer processes measured signals for fault detection, fault ride-through support, and transition to islanded operation, while the inner control loops remain physics-based and deterministic.

4.1.2 Model Identification (incl. Digital Twin)

Physics-informed machine learning methods: The continuous integration of AI into power electronics has paved the way for advanced control, condition monitoring, and predictive maintenance. However, classical data-driven machine learning models often face significant hurdles in practical power electronics applications. The black-box nature of pure deep learning inherently lacks physical interpretability and causality, struggling to ensure safety and reliability. Moreover, these purely data-driven models require massive amounts of high-quality data to learn complex non-linear behaviors, which are rarely available across diverse operating boundaries and varying mission profiles in power electronics. To overcome these limitations, physics-informed machine learning (PIML) has emerged as a promising paradigm [188].

By seamlessly embedding fundamental physical principles, such as Kirchhoff's laws and ordinary differential equations (ODEs) into the learning process, PIML bridges the gap between physics-based analytical models and data-driven methods. The embedded physics effectively regularize the learning space, reducing dependency on exhaustive datasets and drastically improving the data efficiency. Most importantly, PIML enhances model robustness, ensuring physically consistent predictions even when facing unseen operating conditions, thus addressing the reliability constraints crucial for digital twins and prognostics and health management (PHM) in power electronics. A representative application highlighting the potential of PIML is its implementation in parameter estimation [189], which is a core requirement for building accurate digital twins of power converters. Accurate extraction of in-situ component parameters (e.g., inductance, capacitance, and equivalent series resistances) under varying operating conditions is traditionally challenging due to noise, measurement errors, and the requirement for invasive hardware intervention. A novel PIML-based framework has been developed and successfully demonstrated on a DC-DC Buck converter, as shown in Figure 4.2a. In this approach, a deep neural network is coupled with the converter physical dynamic model of the converter. This physics-informed approach demonstrates remarkable robustness. This case exemplifies how PIML unlocks new possibilities for data-light and highly reliable digital twins in power electronic systems.

Look-up tables or functions for non-linear parameters: While discussing the power electronics application, many of the system behaviors corresponding to switching losses, thermal effect as well as the system efficiency are prominently nonlinear in nature and cannot be represented by some



(a) Physics-informed machine learning for parameter estimation in PE systems [189].

(b) Thermal mapping of a power electronics device through look-up table and generative AI [190].

Figure 4.2: Illustrations related to model identification.

functional equations. Hence for the analysis and assessment of such nonlinear models for AI application, it is a common practice to use look-up tables (LUTs) and non-linear functions. The LUTs provide a mapping of input-output correlation based on the precomputed data which is observed experimentally or though simulated studies. It usually tends to assist with rapid real-time evaluation even with limited computation. LUTs are especially beneficial in embedded controllers and AI based optimization where speed and reliability play a critical role.

However, data limitation can be one of the restricting factors for LUTs hence power electronics phenomena are expressed with non-linear functions, that can be represented as continuous characteristics. Expressions such as polynomial for temperature dependent resistance or exponential for diode characterization are some of the prime examples of nonlinear functions for power electronics dynamics representations. In contrast to the lookup tables, non-linear functions operate with less memory and can produce operating points with more continuity, which makes them appropriate for control and system-level simulations.

Further the intelligent and machine learning techniques tend to learn through mapping of nonlinear data, however intelligent models such as Gaussian processes, neural networks, support vector regression have performed better at adopting the non-linear patterns and presents itself as an alternative for LUTs and analytical methods. Moreover, in practice, a hybrid approach is commonly found where the LUTs act as input to train the intelligent model while also considering the physics-based functions. The combination tends to find balance between the accuracy and computation efficiency making AI driven power electronics more reliable and future-proof.

ML-Methods in combination with thermal models (thermal neural network (TNN)): The thermal modeling for power electronics devices, particularly wide-bandgap devices, has been represented through RC network. Wide-bandgap power modules demonstrate non-linear temperature dependencies along with transient behavior, that can be represented by either traditional Foster [191] or Cauer models [192]. However, such models at times struggle to capture the dynamics precisely. Therefore, Machine learning based model have factored in to improve the predictability of the traditional methods. Many researchers have explored intelligence techniques when it comes to dynamic power loss variations [193] and non-uniform heating [194]. There has also been some sustainable work with Generative Adversarial Networks for developing a temperature dependent RC framework [190], as illustrated in Figure 4.2b, with an aim to emulate the dynamics of temperature dependent thermal parameters accurately. In summary, merging machine-learning techniques along with thermal

network models presents more adaptive and accurate framework when it comes to predicting device temperatures for a power-electronic systems.

AI-enabled digital twins for power electronics systems: Digital twin frameworks are increasingly investigated for advanced monitoring, optimization, and reliability assessment of power electronic systems. In this context, a digital twin represents a dynamic virtual counterpart of the physical converter that integrates real-time measurements, physics-based electro-thermal models, and data-driven learning algorithms to reproduce system behavior across varying operating conditions. Classical implementations rely on circuit models, semiconductor loss models, and thermal RC networks to capture interactions between electrical variables, switching losses, and junction temperature dynamics. However, accurately identifying model parameters and capturing complex nonlinear couplings between operating conditions, device degradation, and thermal effects remain challenging in practical systems.

Recent advances in AI enable the integration of machine-learning-based surrogate models within digital twin architectures to approximate multidimensional loss surfaces, efficiency maps, and electro-thermal dynamics with significantly reduced computational complexity. Such approaches have been investigated for real-time estimation of semiconductor junction temperatures, converter efficiency prediction, and condition monitoring of power modules [195, 196].

Furthermore, hybrid modelling strategies combining physics-based models with data-driven learning, including physics-informed neural networks, allow the enforcement of physical constraints such as energy balance and thermal dynamics during training, improving model generalization and robustness [197]. These AI-enabled digital twins therefore provide a promising framework for predictive maintenance, reliability-aware operation, and adaptive optimization of next-generation power electronic systems.

4.1.3 State Estimation

Any PE system can be represented as a nonlinear discrete-time state-space model

$$\boldsymbol{x}_{k+1} = f(\boldsymbol{x}_k, \boldsymbol{u}_k) + \boldsymbol{w}_k, \qquad \boldsymbol{y}_k = h(\boldsymbol{x}_k, \boldsymbol{u}_k) + \boldsymbol{v}_k, \tag{4.1}$$

where $\boldsymbol{x}_k \in \mathbb{R}^n$ is the internal state, $\boldsymbol{u}_k \in \mathbb{R}^m$ the known input, and $\boldsymbol{y}_k \in \mathbb{R}^p$ the measured output; $\boldsymbol{w}_k$ and $\boldsymbol{v}_k$ capture model uncertainty and measurement noise. In practice, sensors provide only $\boldsymbol{y}_k$ (often with $p < n$), while $h(\cdot)$ may be nonlinear and distorted by switching ripple and EMI. State estimation therefore reconstructs $\boldsymbol{x}_k$ from the input–output history. Typical tasks in PE are summarized in Table 4.2.

Formally, *filtering* computes the online estimate $\hat{\boldsymbol{x}}_{k|k} = \mathcal{E}(\boldsymbol{y}_{0:k}, \boldsymbol{u}_{0:k-1})$, i.e., the current state from measurements up to time k and the applied inputs. Depending on the application, one may also consider *prediction* ($\hat{\boldsymbol{x}}_{k+\ell|k}$) or *smoothing* ($\hat{\boldsymbol{x}}_{k|N}$ with $N > k$). Classical observers realize $\mathcal{E}(\cdot)$ through a model-based time update with $f(\cdot)$ and a measurement correction driven by the innovation $\boldsymbol{y}_k - h(\hat{\boldsymbol{x}}_{k|k-1}, \boldsymbol{u}_k, k)$. ML-assisted estimators retain this structure but learn parts of f, h, and/or the update law to reduce modeling errors and nonstationary uncertainties.

In *open-loop* estimation, the state is reconstructed without innovation feedback, e.g., $\hat{\boldsymbol{x}}_k = g(\boldsymbol{y}_{0:k}, \boldsymbol{u}_{0:k})$, where $g(\cdot)$ is an (identified) model. Open-loop estimators offer low computational cost and simple deployment, but they remain sensitive to model mismatch, drift, and nonstationary disturbances,

e.g., switching ripple, EMI and aging. In contrast, *closed-loop* estimators feed back the innovation to drive convergence, e.g., through the following possibly time-varying observer structure

$$\hat{\boldsymbol{x}}_{k+1|k} = f(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{u}_k, k), \qquad \hat{\boldsymbol{x}}_{k+1|k+1} = \hat{\boldsymbol{x}}_{k+1|k} + \boldsymbol{L}_{k+1}\big(\boldsymbol{y}_{k+1} - h(\hat{\boldsymbol{x}}_{k+1|k}, \boldsymbol{u}_{k+1}, k{+}1)\big)\,, \tag{4.2}$$

covering classical Luenberger-type designs and (extended/unscented) Kalman filters as prominent instances [198, 199, 200]. For PE operation, this distinction naturally extends to *mutual state and model estimation*: unknown or slowly varying parameters $\boldsymbol{\theta}$, such as L, C, equivalent series resistance (ESR), or loss and thermal parameters, are estimated jointly with the states by augmenting $\boldsymbol{x}$, i.e., $\boldsymbol{x}_k^a = [\boldsymbol{x}_k^\top, \boldsymbol{\theta}_k^\top]^\top$, or by using dual/adaptive schemes [201, 202, 203]. This parameter tracking can serve as a health indicator for predictive maintenance (cf. subsection 4.1.5) and fault detection (cf. subsection 4.1.4). Because estimation quality ultimately depends on the fidelity of $f(\cdot)$, $h(\cdot)$, and their parameters, data-driven model identification and digital twins are discussed separately in subsection 4.1.2.

Table 4.2: Typical State-Estimation Use Cases in PE Systems (Incl. Drives) and Representative Literature Using ML-Free Methods

Use category (incl. typical states)	Typical online purpose	Examples
Sensorless electric drives ($\boldsymbol{\theta}_e$, ω, flux/electromotive force (EMF)-related states)	Eliminate mechanical sensors or provide back-ups for sensor failures	[204, 205, 206]
Converter electrical states (inductor currents, capacitor voltages)	Reduce sensor count (virtual sensing) and compensate load impacts	[207, 208, 200, 209]
Grid-interface states (grid voltage magnitude/angle/frequency)	Synchronization and control in distorted/unbalanced grids	[210, 211, 212, 213]
Thermal states (e.g., junction or electric machine temperatures)	Thermal derating and lifetime-aware operation w/o intrusive sensors	[214, 215]

Classical observers offer structure, interpretability, and convergence guarantees when accurate models are available, but practical PE systems exhibit operating-point-dependent parasitics, mode-dependent behavior, sensor imperfections, and aging-induced drift that remain difficult to model over the lifetime. Data-driven and hybrid ML approaches therefore reduce reliance on highly accurate expert models and manual tuning, improve robustness to mismatch and nonstationary uncertainty, and enable *virtual sensing* from limited measurements; representative examples are summarized in Table 4.3. In safety-critical applications, engineers usually deploy ML as an *assistant* to a physics-based estimator, for example to learn gains, noise statistics, or residual dynamics, rather than replace the estimator end-to-end [216, 217].

A recent ML-based state-estimation example is shown in Figure 4.3. In magnetic cores for PE converters, transient B–H trajectories form a nonlinear dynamical system with hidden internal states: although $H(t)$ and $B(t)$ are measurable, the underlying domain configurations, pinning effects, and rotation processes are not [247, 248, 249]. As a result, the same instantaneous B can map to different H values depending on excitation history, so accurate models require internal memory variables.

Microscopic white-box approaches such as micromagnetics and domain theory can explain this transient ferromagnetic behavior, but transferring them to macroscopic core-sample trajectories remains a computationally intensive multiscale problem that scales poorly to device-level simulation [249, 250]. Recurrent or hybrid neural hysteresis surrogates therefore offer an attractive alternative because they learn latent memory dynamics directly from measured loops and still support fast inference for repeated use in magnetic-component and converter-design workflows [244, 243, 242].

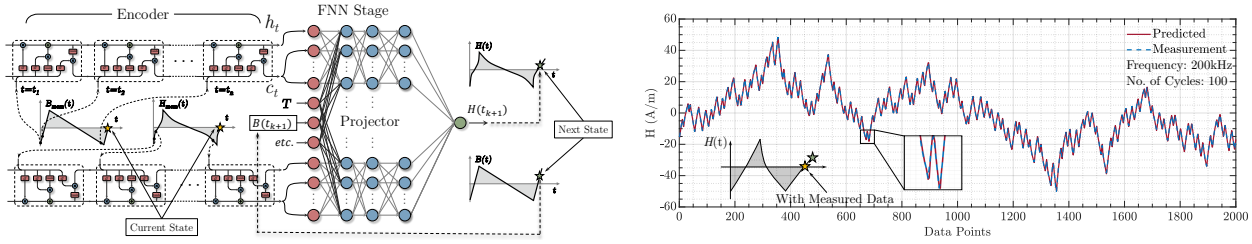
Table 4.3: ML-Assisted and ML-Based State-Estimation Use Categories in Power-Electronic Systems and Representative Literature

Use category / estimated quantities	Typical ML role in the estimator loop	References
Sensorless drives: rotor angle/speed	Pure black-box or ML-assisted estimation, e.g, ANN tunes observer parameters	[218, 205, 219, 206, 220, 221, 222]
Sensorless drives: rotor flux	Black-box or ML-assisted estimation of unmeasured induction-machine rotor flux	[223, 224]
Virtual sensing of currents and voltages	Open-loop state estimation combining physics-informed priors with ML models	[225, 189, 226, 227]
Current/voltage denoising during transients	Recursive ANN denoising for EMI-corrupted transient signals, often with less delay than classical filters	[228, 229, 230]
Grid-connected converters: voltage/harmonic monitoring	Infer unmeasured voltage/harmonic indices (e.g., total harmonic distortion (THD)-related quantities) from local converter measurements	[231, 232, 233, 234]
Temperature estimation in converters and drives	ANN submodels embedded in lumped-parameter thermal networks to capture parasitics, nonlinear parameters, and loss effects	[235, 236, 237, 238]
Magnetic state estimation in inductive components (transient B–H curve models)	Open-loop dynamic ML models for transient B–H trajectories in magnetic components	[239, 240, 241, 242, 243, 244]
ML-aided filtering (generic, but PE-applicable): gain/noise learning, latent-space filtering	Keep a Kalman/observer scaffold, but learn $\boldsymbol{L}_k$, $\boldsymbol{Q}/\boldsymbol{R}$, residual dynamics, or latent encoders	[216, 217, 245, 246]

4.1.4 Fault Detection

Fault detection describes the recognition of discrete abnormal events. This distinguishes it from predictive maintenance (subsection 4.1.5), which focuses on gradual degradation phenomena. Despite the difference in timescale, both rely on similar signals and feature pipelines. Therefore, practical system architectures often implement shared signal preprocessing, with high-priority fault detection operating at short latency and degradation tracking executed in background tasks.

In power-electronic systems, fault detection aims to detect abnormal conditions at an early stage, localize the most probable source (e.g., device, sensor, gate drive, passive components, interconnect, load, or grid), and trigger an appropriate reaction such as warning, derating, reconfiguration, or shutdown [251]. While this objective appears conceptually straightforward, product operation imposes strict constraints. In contrast to laboratory diagnosis, sensing capability is limited, computa-



(a) LSTM-based estimator approach. **(b)** Exemplary test result in the time domain.

Figure 4.3: ML-based open-loop state estimation for transient B–H trajectories within magnetic components (source: [240]).

tional resources are bounded, and false trips must be minimized without compromising protection speed.

To effectively select detection strategies, a systematic understanding of fault types is essential, as illustrated in Figure 4.4. Power-electronic converters are exposed to a wide range of electrical, thermal, environmental, and control-related faults as reported extensively in the literature [252, 253]. The major fault categories relevant for converter operation are summarized in Table 4.4.

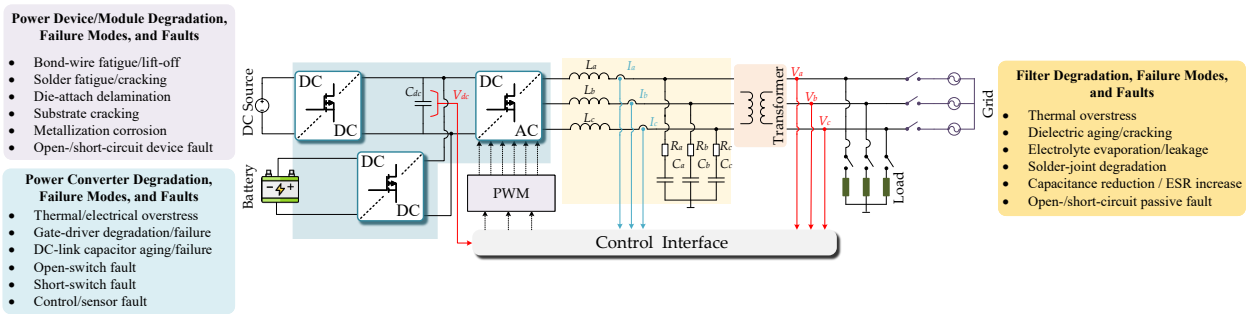


Figure 4.4: Fault layout for grid-connected system (inspired by [252]).

Table 4.4: Fault Classifications for Power Electronics Converter [254, 255, 256, 257]

Fault classification	Explanation	Underlying factor	Major consequences
Semiconductor device faults	Failures in power switches	Short circuits, bond-wire lift-off, overheating, gate failure	Shoot-through, converter shutdown, device destruction
Capacitor faults	Failure of DC-link or filter capacitors	ESR increase, capacitance loss, insulation failure	DC-bus ripple, overvoltage, reduced lifetime
Magnetic faults	Issues in inductors and transformers	Winding shorts, insulation breakdown, core saturation	Overcurrent, distortion, excessive heating
Gate driver & control faults	Faults in switching commands	Missing pulses, controller failure	Uncontrolled switching, shoot-through, protection trips
Bus faults	Irregularities in interconnection wiring	Overvoltage, grounding issues, insulation breakdown	System shutdown, semiconductor stress
Input connection faults	Faults from grid or supply	Over/under-voltage, phase loss, surges	Component stress, converter malfunction
Output connection faults	Faults at the load side	Short circuits, overload, ground faults	Overcurrent, converter shutdown
Conditional faults	Environmental and operational stresses	Temperature extremes, vibration, humidity	Long-term degradation, intermittent failures
System-level faults	Combined or cascading faults	Protection failure, EMI, communication loss	Multi-component damage, instability

These fault classes directly influence the choice of detection methodology. In general, abrupt electrical faults (e.g., short circuits, missing gate pulses) require fast detection mechanisms embedded in protection layers, whereas parametric or environmental faults often require trend-based or model-based monitoring strategies. In the following subsections, we discuss various methods and associated challenges.

Model and Signal-based Methods

These approaches rely on residuals or diagnostic features that remain small during healthy operation and deviate in the presence of faults [258]. In practice, three recurring patterns are observed:

- *Residual generation* using observers (e.g., Luenberger- or Kalman-like observers) or parity relations between measured currents/voltages and expected values.
- *Feature extraction* from electrical signals, including sequence components, harmonic indicators, spectral energy in selected frequency bands, wavelet coefficients, switching transient features (e.g., $\mathrm{d}i/\mathrm{d}t$, $\mathrm{d}v/\mathrm{d}t$ proxies), and dc-link ripple characteristics.
- *Consistency checks* between redundant or physically related signals (e.g., phase currents versus dc-link current balance, commanded versus realized switching states).

The primary advantages of these methods are interpretability, low data requirements, and easier verification and validation within established industrial development and safety-assurance processes. However, their performance depends on model accuracy and operating conditions, which may limit robustness under parameter drift or strongly varying load profiles.

AI-based fault detection

AI-based methods are particularly beneficial when fault signatures are subtle, multi-factorial, or masked by noise and operating variability [259, 260]. In industrial implementations, three categories are most relevant:

- *Supervised classification*, applied when labeled fault data are available (e.g., from accelerated aging tests, hardware in the loop (HiL) setups, or field return analysis). Lightweight classifiers such as shallow neural networks, tree-based models, and support vector machines (SVMs) are typically preferred for embedded targets.
- *Unsupervised or one-class anomaly detection* for rare or previously unseen faults. Methods such as autoencoders, one-class SVM, density estimation models, or clustering in feature space are trained using healthy data and detect deviations from learned normal behavior.
- *Temporal models* for incipient faults. Sequence-based models, such as RNNs (e.g., LSTM, GRUs) and temporal CNNs, operate on sliding windows to identify trends and change points rather than isolated deviations.

Irrespective of the selected algorithm, feature robustness across operating points and real-time computability remain the primary design constraints.

Hybrid concepts, practical challenges, and industrial relevance

Given the complementary strengths of analytical and data-driven approaches, hybrid architectures are frequently adopted. A common structure combines a fast and interpretable physics-based residual with an AI-based decision layer that adapts thresholds, compensates operating-point dependencies, or fuses multiple weak indicators. Related ideas are already visible in the literature, for example in converter fault diagnosis schemes that combine signal- or transform-based feature extraction with lightweight classifiers such as multilayer perceptrons or random forests for real-time inverter fault detection and localization [261, 262]. Embedding physics-informed constraints such as monotonicity, energy consistency, or thermal causality further improves robustness and generalization.

From an industrial perspective, deployment challenges include class imbalance (faults are rare), domain shift between laboratory and field data, sensor limitations (bandwidth, quantization, drift), and the definition of appropriate reaction strategies. Consequently, evaluation metrics in product operation should emphasize *time-to-detect* and *false-trip rate* under representative duty cycles rather than relying solely on offline accuracy.

4.1.5 Predictive Maintenance

PdM for power electronics estimates the *health state* and *RUL* of critical components using condition data collected during operation. In contrast to fault detection, PdM targets *slow degradation dynamics*, where parameter drift evolves over mission profiles defined by load, thermal cycling, humidity, vibration, and switching stress [263, 264, 265].

Dominant aging mechanisms in power electronics: Wear-out mechanisms are primarily driven by thermo-mechanical stress and electrical overstress. These include interconnect degradation (bond-wire lift-off, solder fatigue), packaging and thermal interface material (TIM) aging (increase in thermal resistance), gate oxide degradation, and passive component aging (e.g., dc-link capacitor ESR increase and capacitance loss). Since degradation strongly depends on mission profiles, PdM must operate on *cycle-based stress accumulation* rather than static lifetime assumptions.

Health indicators and degradation observables: In practical systems, degradation is inferred indirectly through measurable proxies that evolve over time. These include electrical signatures (e.g., on-state voltage drift, switching transient variation), thermal responses (e.g., increased junction temperature for constant losses), and operational statistics (e.g., derating frequency, stress cycle histograms). The key challenge is decoupling degradation-induced variations from normal operating-point dependencies.

This relationship between measurable indicators, hidden degradation states, and decision layers is illustrated in Figure 4.5, highlighting the multi-layer inference process required for reliable PdM.

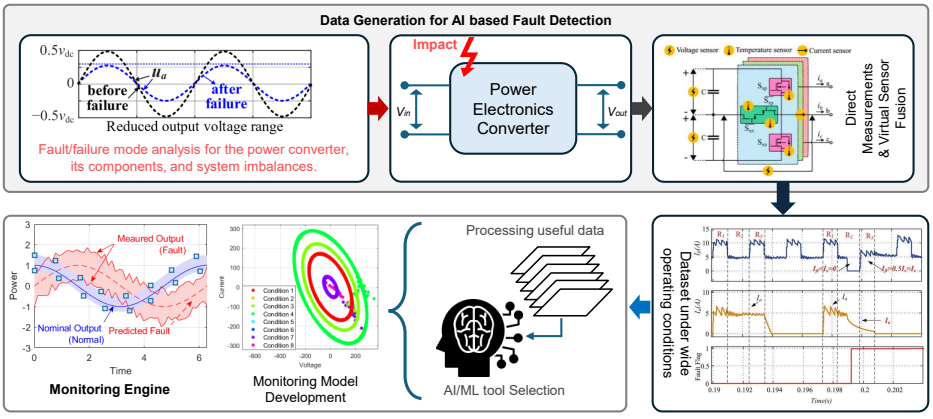


Figure 4.5: Predictive maintenance architecture for power-electronic systems.

Predictive maintenance methodologies: PdM approaches can be broadly categorized into physics-based and data-driven methods, both of which rely on consistent mapping between stress, degradation, and observable features [19].

(1) *Physics-based lifetime modeling:* Physics-based PdM computes accumulated damage from stress cycles using electro-thermal models and empirical lifetime laws. This approach provides inter-

Table 4.5: Mapping Between Degradation Mechanisms and Observable Indicators in Power Electronics [266, 267, 268]

Component / failure mode	Dominant stress driver	Observable indicator	Measurement Domain
Bond-wire fatigue	Thermal cycling (ΔT_j)	Increase in on-state voltage $V_{CE(on)}$ / $R_{DS(on)}$	Electrical
Solder / die attach degradation	Power cycling, thermo-mechanical stress	Increase in thermal impedance Z_{th}	Electro-thermal
DC-link capacitor aging	Ripple current, temperature	ESR increase, capacitance drop, DC-link ripple	Electrical
Magnetic insulation degradation	Thermal + electrical stress	Core losses, harmonic distortion	Electromagnetic

pretability and direct linkage to failure mechanisms but requires accurate estimation of internal states such as junction temperature and stress cycles [263, 269].

(2) *AI-based health and RUL estimation:* Data-driven approaches learn mappings from indicators to health state or RUL. These include regression models, temporal sequence models, and probabilistic estimators capable of capturing degradation trends under varying operating conditions [265]. The uncertainty in the RUL should be explicitly considered, where statistical and stochastic models [270] can be applied together with AI methods for probabilistic predictions. Due to limited failure data, hybrid training strategies such as transfer learning and semi-supervised learning are commonly required [19].

Digital twins and hybrid PdM concepts: Hybrid approaches combine physics-based models with AI to balance interpretability and adaptability. In this structure, physics models estimate stress and internal states, while AI compensates for modeling errors or estimates unmeasured parameters. This improves generalization across converter variants while reducing dependence on large labeled datasets.

Table 4.6: Comparison of PdM Modeling Paradigms for Power-Electronic Systems

Approach	Model structure	Key limitation	Industrial suitability
Physics-based	Electro-thermal + lifetime models	Requires accurate stress/state estimation	High (interpretable, certifiable)
Data-driven (AI)	Feature $\rightarrow$ RUL mapping	Data scarcity, limited extrapolation	Medium (requires validation)
Hybrid (digital twin)	Physics + ML correction	Increased integration complexity	High (balance of accuracy and robustness)

As summarized in Table 4.5 and Table 4.6, effective PdM requires consistent alignment between degradation physics, observable features, and model structure.

Industrial relevance and open challenges: In industrial deployment, PdM must remain robust against sensor drift, missing data, and evolving operating conditions. A key practical challenge is not model accuracy but threshold calibration and decision robustness under uncertainty. Therefore, PdM systems require continuous validation, adaptive thresholding, and integration with supervisory control to ensure safe and reliable operation over the product lifetime.

4.2 Industrial Deployment Challenges

4.2.1 Cybersecurity & Data Privacy

AI-enabled functions in product operation increase the cyber attack surface by introducing additional data flows (logging, connectivity, updates) and creating high-value assets such as trained models, feature pipelines, and condition data. Consequently, cybersecurity must be co-designed with performance and reliability requirements rather than treated as an add-on [271]. For power-electronic systems, the primary security objectives remain *availability*, *integrity*, and *confidentiality*, but their realization is complicated by real-time constraints and limited embedded resources.

The integration of AI introduces tightly coupled cyber-physical vulnerabilities. Manipulation of sensor signals (e.g., current, voltage, temperature) can propagate through observers and learning-based estimators, leading to incorrect control or protection actions. Similarly, communication interfaces (fieldbus, Ethernet, wireless, cloud) expose the system to replay and injection attacks if authentication and encryption are insufficient [272]. Beyond data channels, AI-specific assets such as model parameters and training datasets create new risks, including model tampering and data poisoning, which can systematically degrade detection accuracy or bias decision boundaries.

To address these challenges, security must be enforced across the full lifecycle of AI components. In practice, AI artifacts (models, feature definitions, calibration data) should be handled equivalently to safety-critical firmware, ensuring integrity, authenticity, and traceability [273]. This includes secure boot mechanisms, signed model deployment, authenticated communication channels, and runtime validation of inputs and outputs. In particular, bounding and plausibility checks on AI outputs are essential to prevent propagation of erroneous decisions into control loops. Furthermore, update mechanisms must incorporate rollback protection and strict version control to avoid unintended downgrade attacks.

Table 4.7: AI-Induced Vulnerability Propagation Paths in Power-Electronic Systems

Injection point	Affected internal variable	Propagation mechanism in AI/control loop	Mitigation strategy
Analog sensor channel (analog-to-digital converter (ADC) input)	State estimates (i_d, i_q, T_j)	Bias/noise alters observer residuals → incorrect state reconstruction → degraded control law	Signal plausibility checks, redundancy, observer consistency monitoring
Digital communication bus (CAN/Ethernet)	Supervisory references, setpoints	Replay/injection modifies reference trajectory → mismatch with plant dynamics → instability or constraint violation	Authenticated communication, sequence counters, anomaly detection
ML inference layer (weights/activations)	Learned mappings (e.g., loss/thermal models)	Weight perturbation or adversarial input → distorted nonlinear mapping → incorrect compensation or prediction	Model signing, runtime bounds, adversarial robustness testing
Training/update pipeline	Model parameters and decision boundaries	Poisoned dataset → biased feature distribution → long-term degradation of detection accuracy	Dataset validation, drift detection, controlled retraining procedures
Feature extraction stage	Derived features (harmonics, RMS, gradients)	Feature distortion → misclassification or anomaly masking in AI models	Robust feature design, normalization checks, redundancy in feature space

The relationship between these threat vectors and their propagation paths across sensing, inference, and actuation layers is summarized in Table 4.7 and illustrated conceptually in Figure 4.6. As shown,

disturbances introduced at the data level can cascade through AI models and ultimately affect physical system behavior, highlighting the need for cross-layer protection mechanisms.

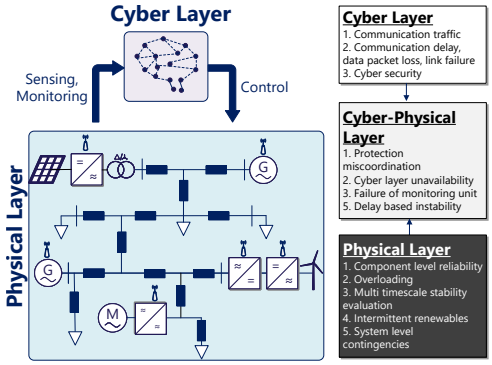


Figure 4.6: Conceptual illustration of cyber-physical attack propagation in AI-enabled power-electronic systems, showing interaction between sensing, AI inference, and control layers [274].

In addition to security, data privacy must be considered due to the increasing use of operational data for analytics and predictive maintenance. Such data may reveal sensitive information about usage patterns or system operation. Therefore, privacy-preserving design principles should be embedded in the data pipeline, including local feature extraction (edge processing), minimal data transmission, and controlled data retention. When fleet-level learning is required, approaches such as federated learning can reduce raw data exchange; however, they still require robust authentication and protection against malicious updates.

Overall, cybersecurity and data privacy directly influence system reliability and safety. In AI-enabled power-electronic products, compromised data or models can lead not only to incorrect monitoring results but also to unsafe control actions, making secure and privacy-aware design a fundamental requirement for deployment.

4.2.2 Functional Safety / Production Deployment

The integration of ML-models into embedded and safety-critical environments can present a significant challenge for traditional certification and safety standards like ISO 26262. Most standards were primarily designed for deterministic, rule-based software, which differs fundamentally from the data-driven nature of ML-models.

The requirements for functional safety that a ML model must fulfill depend highly on the environment where the model is deployed and the task of the model. For example, a supplementary virtual temperature sensor is in no way relevant to safety. In the worst case of a failure or exceptional behavior the main system is rarely influenced. On the other hand a model that is integrated into the current controller of an electric vehicle inverter is highly safety-critical as it might influence the acceleration of the vehicle.

From a regulatory perspective, emerging frameworks such as the European Union (EU) AI Act classify most automotive AI applications, particularly those influencing actuation or safety-relevant decisions, as *high-risk systems*. This implies additional requirements on transparency, traceability, dataset governance, and post-deployment monitoring, which extend beyond traditional functional safety standards and must be integrated into the development lifecycle.

To address the non-deterministic nature of ML models, explainability, and uncertainty quantification become key enablers for safety assurance. In practice, embedded power-electronic applications often

employ compact models (e.g., shallow neural networks or regression-based approaches), which allow partial interpretability and bounded behavior. This facilitates verification, while uncertainty metrics (e.g., confidence bounds or residual monitoring) enable runtime detection of unreliable predictions.

In current industrial practice, safety-critical deployments typically avoid online self-learning models, as continuously changing model parameters violate certification assumptions. Instead, fixed (offline-trained) models with controlled updates are preferred, ensuring traceability and repeatability of behavior across the product lifecycle.

Recent industrial approaches emphasize a unified safety lifecycle for AI systems, combining ISO 26262, ISO 21448 (safety of the intended functionality (SOTIF)), and emerging standards such as ISO/PAS 8800. These frameworks extend safety concepts to include robustness, resilience, and operational monitoring, addressing failure modes originating from data insufficiencies and distribution shifts rather than hardware faults.

Building upon these mechanisms, safety assurance requires embeddingML components into a hierarchical safety architecture with explicit monitoring and fallback strategies. In such systems, ML outputs are continuously supervised, and deviations from expected behavior trigger deterministic safety mechanisms.

Table 4.8: Failure Propagation and Safety Mechanisms for ML Components in Power-Electronic Systems

Failure source	Affected variable	System-level effect	Mitigation mechanism
Training data bias	Estimated states (e.g., T_j, flux)	Systematic error → incorrect derating or protection	Dataset validation, safety margins
out-of-distribution (OOD) input / distribution shift	ML model output	Unbounded prediction → instability or constraint violation	OOD detection, runtime monitoring
Quantization / implementation error	Model inference output	Deviation from trained behavior → degraded control accuracy	Toolchain validation, error bounding
Sensor anomaly	Input features	Distorted feature space → misclassification or estimation error	Input plausibility checks, redundancy

The interaction between ML components, safety monitors, and fallback mechanisms is illustrated in Figure 4.7. The architecture highlights how safety constraints are enforced across multiple layers, ensuring that ML-induced deviations do not propagate unchecked into the physical system.

Overall, functional safety for ML-based systems shifts from purely deterministic verification towards a combination of bounded model behavior, runtime monitoring, and system-level safety enforcement. This aligns with emerging safety concepts where robustness and resilience are treated as primary design objectives in AI-enabled embedded systems.

4.2.3 Embedded Hardware Requirements

Optimization is crucial for ML-models deployed to embedded hardware, as computational resources are tightly constrained by cost pressure. A typical target environment is a microcontroller running a real-time operating system that schedules tasks at fixed periodic intervals. The latency budget available for model inference therefore depends directly on which task hosts the model: a neural network embedded in the fast control loop of a power inverter faces far stricter time and memory limits than one running in a slower supervisory task on the same device.

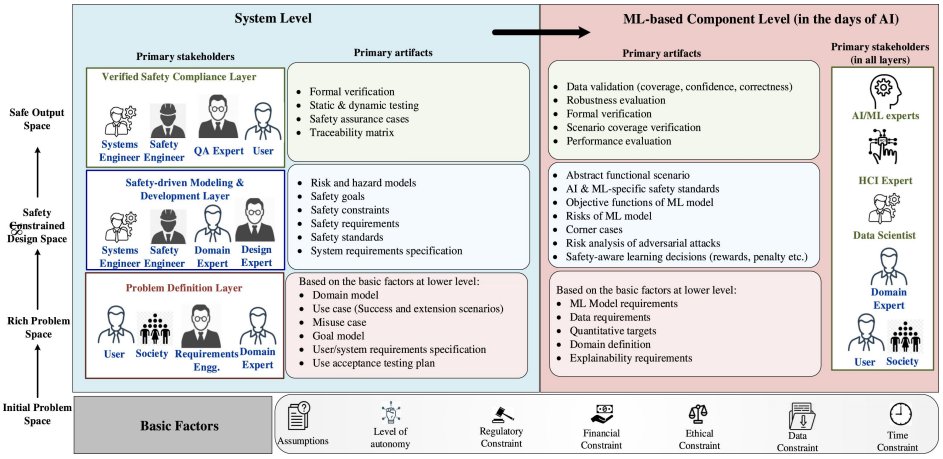


Figure 4.7: Layered safety architecture for ML-enabled power-electronic systems, showing interaction between ML inference, runtime monitoring, and fallback control mechanisms.

The most prominent optimization is model compression, either during training or as post-processing step. The aim is to reduce the memory footprint and the inference time with acceptable impact on model accuracy [275]. Table 4.9 summarizes popular techniques:

Table 4.9: Model Compression Techniques for Edge Deployment

Method	Description	References
Architecture search	During neural architecture search (NAS), or more specifically, a multi-objective strategy trades model performance against hardware constraints (e.g. parameter count or multiply-accumulate (operation) (mac) operations), exposing a Pareto front from which an architecture is selected.	[276]
Pruning	Weights or neurons with negligible output contribution are removed. *Structured* pruning (whole filters or layers) yields direct speedups on standard hardware; *unstructured* pruning requires sparse-tensor support to realize gains.	[277]
Quantization	ML frameworks default to 32-bit floating-point (`float32`) arithmetic. Reducing precision to 16-bit floats (`float16`), 8-bit integers (`int8`), or even 4-bit integers shrinks the memory and storage footprint proportionally and accelerates arithmetic on fixed-point hardware. Two variants are common: post-training quantization (PTQ) converts a trained model, while quantization aware training (QAT) simulates quantization noise during training.	[278]
Knowledge distillation	A compact *student* network is trained to reproduce the output distribution of a larger, more accurate *teacher* model rather than learning from hard labels alone. The teacher's *soft* predictions carry richer gradient information and allow the student to reach accuracy levels otherwise unattainable at its parameter budget.	[279]
Activation functions	Embedded environments often do not provide the same instruction set as desktop workstations. Especially common activation functions as `tanh` or `sigmoid` may only be available as approximations. The target platform needs to be considered during training.	[280]

AI models deployed in a hardware must be robust against memory errors, power interruptions, and harsh environments such as vibration, temperature variations, and electromagnetic interference. These conditions impact both hardware selection with built-in error correction, timers, and model design, which may require additional validation or redundancy to ensure reliable operation.

In recent years major companies in the microcontroller (MCU) industry started offering accelerator cores in their chips. The individual companies use their respective brand names like neural processing unit (NPU) (Texas Instruments), Neural-ART (ST) or NNLite (Infineon), but share a mac array optimized for the matrix–vector operations that dominate neural network (NN) inference [281]. These accelerators operate exclusively on fixed-point arithmetic (`INT8/INT4`). Quantisation is therefore both a compression technique and a hard prerequisite for hardware-accelerated deployment. Complementary to hardware innovation, MCU manufacturers support development with specialized toolchains. For example, ST's *X-Cube-AI* converts open neural network exchange (ONNX), TensorFlow Lite, and Keras models to optimised C for STM32 targets [282].

4.2.4 Data Requirements and System Excitation

ML-based models, observers, and controllers are rapidly entering PE systems, yet data quality still receives less attention than model architectures or training algorithms [235, 283, 284]. This gap is critical because PE dynamics span several decades of time scale, from switching transients to slow thermal behavior; sensor bandwidth, synchronized sampling, anti-aliasing, resolution, and noise level must therefore match the phenomenon of interest, since learning cannot recover information lost in the measurement chain [285, 286]. Accordingly, *good data* must be informative, representative, well distributed, and physically plausible: they must cover the admissible operating domain, relevant transients, boundary regions, disturbances, and device variability while minimizing train–deployment mismatch, because black-box ML methods degrade rapidly under extrapolation and distribution shift [4, 287].

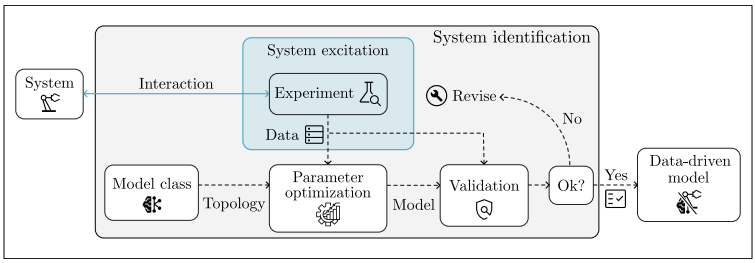


Figure 4.8: Obtaining informative data by system excitation as the obvious prerequisite for any ML approach (inspired by [283]).

This makes design of experiment and system excitation a first-class problem – cf. Figure 4.8. Beyond classical persistency-of-excitation and optimal-experiment-design concepts, nonlinear and safety-constrained PE systems require coverage-oriented or uncertainty-guided excitation strategies that deliberately probe informative regions under hard limits on current, voltage, temperature, EMI, cost, and experimental time [288, 289, 290, 291, 292, 293].

System excitation selects an input policy that drives the system through *safe but informative* operating regions to generate data that maximize later model quality. In a simplified long-horizon setting, neglecting the explicit history state, this reads

$$\pi^\star \in \arg\max_{\pi} \ \mathcal{L}(X,U) \quad \text{s.t.} \quad x_{k+1} = f(x_k, \pi(x_k)),\ x_k \in \mathcal{X},\ \pi(x_k) \in \mathcal{U},\ k = 0,\dots,T-1, \tag{4.3}$$

with state trajectory $X = (x_0, \dots, x_T)$, input trajectory $U = (u_0, \dots, u_{T-1})$ (represented by policy π), and $T \in \mathbb{N} \gg 1$. Above, (4.3) formulates system excitation as an optimal-control problem: $\mathcal{L}$ quantifies rollout informativeness, while $\mathcal{X}$ and $\mathcal{U}$ enforce safe operation. For white-box and grey-box models, informativeness is typically derived from the Fisher information matrix [284, 289];

for black-box models, it is often defined through space-filling metrics that shape target state and input distributions [290, 293, 292]. Solving (4.3) remains a chicken-and-egg problem, because optimal excitation requires a prediction model that the excitation must first identify. In practice, one therefore proceeds iteratively: collect a data batch, update the model, redesign the excitation, and repeat. Data demand further increases because safety-critical edge cases are difficult to measure, internal states are only partially observable, and tolerances, aging, and platform variation broaden the relevant operating domain. Transfer learning can reduce effort across product generations, but it cannot compensate for fundamentally uninformative or misaligned source data [294]. Where ground truth is required, especially for fault or anomaly detection, data annotation and labeling become additional bottlenecks because they require domain expertise and rare events are frequently sparsely represented.

4.3 Future Research Outlook

Over the next decade, academic research on AI for power electronics will transcend conventional deterministic modeling. This evolution will be driven by three tightly coupled megatrends, aiming to bridge the gap between algorithmic capabilities and the stringent reliability demands of industrial systems:

- *Trustworthy uncertainty-aware AI:* The field is shifting from deterministic black-box prediction toward calibrated, decision-aware uncertainty quantification. In critical operation, AI systems will not only predict RUL or monitor condition indicators, but also provide rigorous confidence intervals. Under distribution shifts, such as new operating regimes or component aging, these uncertainty bounds can drive adaptive safety margins, uncertainty-aware derating, and robust fallback strategies. AI outputs thus evolve from point estimates into operationally actionable information that protects system integrity [295, 296, 297, 298].

- *Foundation models for generalizable representation and transferability:* Pretrained representations across time series, spatial graphs, and multi-physics domains are emerging as a key lever against the scarcity of labeled failure data in power electronics. Through cross-topology transfer learning, foundation models trained on standard converter architectures, such as fundamental DC-DC topologies, can be adapted to more complex systems with few-shot fine-tuning. This shift can break down data silos across operating regimes and improve the scalability of digital twins, diagnostics, prognostics, and data-efficient adaptation where labeled failure data remain scarce [299, 300, 24, 301, 302, 303].

- *Agentic AI for supervisory orchestration and human-in-the-loop workflows:* While hard real-time control, such as nanosecond-level gate switching, will remain governed by formal verification, ultra-low latency constraints, and physics-based safety guardrails, AI at the supervisory layer will become increasingly autonomous. Language-enabled, tool-using agents will orchestrate complex engineering and task-management workflows [304, 305, 306, 307, 29]. In fleet-level predictive maintenance for offshore wind converters, for example, an agentic framework could detect early DC-link capacitor degradation, identify low-wind maintenance windows via weather APIs, query enterprise resource planning systems for spare-part availability, and generate dispatch schedules for technician review and authorization.

Taken together, these trends suggest a transition from task-specific AI toward trustworthy, transferable, and partially autonomous hybrid intelligence for operational power electronics, with the

strongest academic impact expected at the interface of physics-based modeling, probabilistic decision support, and supervised human insights.

4.4 Conclusion

Across all considered use cases, the main industrial opportunity of AI in power electronics lies not in replacing established engineering models, but in augmenting them where modeling effort, parameter drift, or multidimensional nonlinearities limit conventional approaches. The most promising path is therefore hybrid AI, bounded, monitorable, and physics-aligned methods that can be embedded into existing product architectures while meeting requirements on real-time capability, safety, cybersecurity, and lifecycle robustness.

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